Hierarchical One-Class Detection for Hyperspectral Image Classification With Background

Chein-I Chang, *Life Fellow, IEEE*, Chia-Chen Liang, Pau-Choo Chung, *Life Fellow, IEEE*, and Peter Fuming Hu

***Abstract*— Hyperspectral image classification (HSIC) has received considerable interest in recent years where most techniques are developed to classify images with background (BKG) removed by ground truth (GT). Unfortunately, in real scenarios, obtaining complete BKG knowledge is generally infeasible. Accordingly, HSIC performed with no BKG (HSIC-NB) is not realistic. Most importantly, many techniques claim to work well for HSIC-NB but perform poorly with BKG included. This article investigates issues arising from BKG in HSIC and further presents a new approach to HSIC with BKG (HSIC-B), called one class detection (OCD), which is based on the well-known hyperspectral subpixel detection technique, constrained energy minimization (CEM). In order for OCD to perform multiclass classification, OCD is further extended to hierarchical OCD (HOC) which is particularly designed to classify multiple classes in a hierarchical tree where each layer uses an iterative kernel CEM (IKCEM) or an iterative kernel target-constrained interference-minimized filter (IKTCIMF) to detect one class at a time for classification. Since M classes are classified by OCD in $M-1$ layers in a hierarchical tree, a new concept of class classification priority (CCP) derived from CEM is specifically designed to rank all the classes along the tree in a prioritized order according to their CCP scores. The experimental results demonstrate that hierarchical OCD (HOCD) works well and performs significantly better than many existing HSIC-NB methods at the expense of slightly reduced classification accuracy compared to HSIC-N methods.**

***Index Terms*— Class classification priority (CCP), hierarchical one-class detection (HOCD), hyperspectral image classification with BKG (HSIC-B), HSIC with BKG removed (HSIC-NB), iterative kernel constrained energy minimization (IKCEM), iterative kernel target-constrained interference-minimized filter (IKTCIMF), iterative random training sampling 3-D convolutional neural network (IRTS-3DCNN), one class classification (OCC), one class detection (OCD).**

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NOMENCLATURE

BKG	Background.
CCP	Class classification priority.
CEM	Constrained energy minimization.
DL	Deep learning.
FSFR	Feature selection based on feature reconstruction.
FSSR	Feature selection based on sample reconstruction.
GT	Ground truth.
HOCD	Hierarchical one class detection.
HSIC	Hyperspectral image classification.
HSIC-B	Hyperspectral image classification with BKG.
HSIC-NB	Hyperspectral image classification with no BKG.
ICEM	Iterative CEM.
IGLPN	Iterative Gaussian Laplacian network.
IKCEM	Iterative KCEM.
IRTS-3DCNN	Iterative random training sampling. 3-D convolutional neural network.
ITCIMF	Iterative TCIMF.
KSR	Kernel sparse representation.
KCEM	Kernel CEM.
KTCIMF	Kernel TCIMF.
LCMV	Linearly constrained minimum variance.
OCC	One class classification.
OCD	One class detection.
AA	Average accuracy.
OA	Overall accuracy.
PR	Precision rate.
SVDD	Support vector data description.
TCIMF	Target-constrained interference-minimized filter.

I. INTRODUCTION

HSIC has been one of the main tasks in hyperspectral imaging (HSI) [1], [2], [3], [4], [5], [6] where numerous works have been reported and published each year. Those people who are interested in the literature review are encouraged

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to refer to [1], [2], [3], [4], [5], and [6]. The traditional HSIC is a multiclass classification problem that classifies an unknown data sample $\mathbf{r}$ into one of the known classes, M classes $\{C_m\}_{m=1}^{M}$. However, an issue arises when $\mathbf{r}$ does not belong to any class of $\{C_m\}_{m=1}^{M}$, i.e., $\mathbf{r} \notin C_m$ for any $1 \leq m \leq M$. Under such circumstance, $\mathbf{r}$ is forced to be classified into one of $\{C_m\}_{m=1}^{M}$, in which case it results in a classification error. So, to avoid being trapped into this dilemma, most of the HSIC techniques reported in the literature assume that the BKG has been removed by GT and the case of $\mathbf{r} \notin C_m$ will not occur. As a consequence, these methods produced very high classification rates, which may mislead people to believing that the same high classification accuracy of these methods may also be applied to HSIC-B. Unfortunately, this is not true as shown in [7], [8], and [9].

Generally speaking, HSIC-NB is unrealistic since obtaining the complete knowledge of BKG is very costly if not impossible. This is mainly due to the fact that HSI can uncover many unknown material substances which result in very complicated BKG. So, characterizing BKG is extremely difficult by prior knowledge. In addition, BKG has shown to have significant impacts in target detection [10] as well as in anomaly detection [11]. The same BKG issue is also applied to HSIC but has not received much attention in HSIC.

A. BKG Issues

BKG issues have been overlooked in HSIC over the past years. In general, we can interpret BKG in two ways. One is to interpret BKG samples as uninteresting data samples and treat BKG samples as a separate class. With this interpretation, BKG consists of various multiple objects which may be identified as objects of no interest by prior knowledge or objects that may not be known by GT. Particularly, BKG varies with different applications. In other words, a BKG sample may be uninteresting to one application but could be vital to another. Using the well-known Purdue Indiana Indian Pines test site [12] as an illustrative example, its BKG contains radio tower, railroads, roads which are of no interest for crop classification in agriculture but are crucial objects in land use/cover, and military applications. In this case, without designated specific applications, BKG cannot be simply removed prior to HSIC.

Alternatively, we can also interpret BKG samples as unlabeled data samples that cannot be identified by GT. In this case, BKG contains unlabeled data samples which belong to their own distinct classes but can be wrongly classified into incorrect classes by HSIC-NB. According to this interpretation, BKG is actually made up of data samples that belong to multiple classes but cannot be labeled or identified by GT. When these unlabeled data samples are mixed with labeled data samples, it creates tremendous confusion and a significant effect on classification accuracy, specifically those unlabeled data samples located nearby or on boundaries among classes. Such mixing results in a drastic reduction of classification accuracy. Therefore, if these unlabeled data samples are removed from classification, the mixing and confusion among labeled and unlabeled data samples no longer exist; thus, it makes HSIC-NB perform very well. It explains why most recent HSIC methods can achieve very high classification accuracy because they are actually HSIC-NB methods. This is particularly true for DL HSIC-NB methods. In many practical applications, due to lack of GT, these unlabeled data samples must be included for classification, in which case HSIC must perform HSIC-B but not HSIC-NB.

B. OCC

To resolve HSIC-B OCC [13] was previously developed. It is also known as outlier detection [14], novelty detection [15], concept learning [16], or single class classification [17], [18], [19], [20] depending upon different applications to which OCC has been applied. On many occasions such as landscape and land cover/use, there may be only a single class of interest needed to be classified, while other classes can be considered as uninteresting classes. This is known as OCC problem and it can be performed using either binary classifiers, by treating all other classes as the negative class, or one-class classifiers which only consider the class of interest. The key difference between these two approaches is in their use of training data and the amount of effort needed to produce it. Binary classifiers require an exhaustively labeled training dataset, whereas OCC only needs to be trained using samples just from the class of interest.

More precisely, in OCC, one of the classes, generally referred to as the positive or target class, can be characterized by the training samples, while the other class, referred to as negative or outlier class, has either no training samples or very few of them, or they do not form a statistically representative sample pool of the negative class. That is, either the negative data training samples are absent or available in limited amount. As a result, only one side of the classification boundary can be determined using merely positive data samples. This makes OCC a challenging problem and more difficult than the conventional two-class classification problem. This is because OCC attempts to define a classification boundary around the positive class, such that it accepts as many data samples as possible from the positive class, while it minimizes the risk of accepting the outlier data samples. Since only one side of the boundary can be determined in OCC, it has problem with deciding how tightly the boundary should fit in each of the directions around the data solely based on one class. In addition, it is also harder to select which features should be used to yield the best separation of the positive class from the negative class. For example, in a recent study, Song et al. [21] developed two feature selection methods using sparse representation for OCC of remote sensing images, FSSR and FSFR to show their performances better than existing OCC methods.

C. OCD

As noted above, OCC does not really address BKG issues since BKG data samples are not considered part of negative class. To be more precise, it is not an HSIC-B method. As a matter of fact, according to the authors' best knowledge, only few studies developed for HSIC-B to account for BKG have been reported in the literature, for example, [7], [8], and [9].

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This article introduces a new concept, called OCD as opposed to OCC. It converts an HSIC-B problem to a single or OCD problem where the class to be classified, is considered as a desired signal, whereas all data samples other than data samples in the class of interest are treated as BKG data samples. Specifically, OCD is derived from a well-known subpixel detection technique, CEM developed by Harsanyi [22], which was designed based on the classic minimum variance distortionless response (MVDR) beamformer in adaptive beamforming [23]. It is a widely used optimal beamforming design, which minimizes the variance of the beamformer output. If the noise is additive, then the variance of the captured signals is the sum of the variances of the desired signal and the noise. Hence, MVDR seeks a solution to minimize this sum, thereby mitigating the effect of the noise. In contrast, the well-known MUltiple SIgnal Classification (MUSIC) algorithm is a method for estimating the direction of arrival of signals received by an antenna array [24]. The algorithm inputs the covariance matrix of the signals received from each antenna element, performs an eigen-decomposition, separates the result into signal and noise subspaces, derives an angular spectrum from this information, and finally searches that spectrum for peaks that represent signal arrival angles. So, to make CEM applicable to OCD, the positive and negative classes characterized by OCC can be treated as desired signal and BKG respectively. With this interpretation, instead of treating unlabeled data samples as negative class as OCC does, CEM considers unlabeled data samples as BKG. Since on many occasions there is partial knowledge about undesired targets, to take care of this issue, CEM was further extended to TCIMF [25] which can be annihilated undesired targets signals while suppressing BKG.

D. Differences in Binary Classification and Detection

OCC divides data samples into positive and negative classes and then performs binary classification by treating the positive class as a one-side class of interest and the negative class as uninteresting class which does not include BKG samples. To evaluate binary classification results, the commonly used confusion matrix is used to calculate overall accuracy (OA) and average accuracy (AA).

In contrast to OCC, OCD considers a class of interest as a desired target class and all other data samples as BKG samples and then performs target detection in BKG. To evaluate the effectiveness of target detection, the detection and false alarm probabilities are calculated to plot a 2-D receiver operating characteristic (ROC) curve for performance analysis. So, fundamentally speaking, binary classification and detection are completely different notions where the former classifies data samples into binary classes as opposed to the latter which detects if a data sample is a target signal via a threshold value. For those who are interested in differences between them, we refer details to [26].

E. CCP

It should be noted that CEM is a technique to detect one single signal in BKG. In order to apply OCD to HSIC-B, CEM must be implemented one class at a time. If there are M classes, it will require CEM to implement $M-1$ times. Accordingly, it yields a completely different approach from OCC, called HOCD which employs CEM in a hierarchical tree with $M-1$ layers where each layer is carried out by a single implementation of CEM. Another issue is that CEM requires the prior knowledge of the desired target signature, $\mathbf{d}$. For CEM to be used for classification, the desired signature for a class to be classified should be obtained by either GT or training samples for HSIC. To resolve this issue, IRTS developed by Chang et al. [27] is perfect to fit its needs such as IRTS-CEM for HSIC [28] and IRTS-HSIC [29].

However, there is an issue associated with implementing HOCD, that is, there are $M! = M(M-1)\cdots 1$ ways to construct a hierarchical tree. To solve this issue, the classes processed in the hierarchical layer should be prioritized. This is because when a class is classified in a hierarchical tree, features of the class to be classified in a layer may overlap with the features of the classes to be classified in its previous layers. Therefore, if the similarity between the features of the classified class and the features of the classes already classified in the previous layers is too high, the classification results may be affected. In other words, HOCD should process the most important class first and then the second most important class and so on. To address this class prioritization issue, a concept of class similarity is defined. If the similarity between the classified class and the classes to be classified in the previous layers exceeds the threshold, the class information should be added back to prevent the current class information from being removed. By incorporating class similarity into the HOCD framework, the classification of subsequent layers will not be affected by the classification results in previous layers. This leads to a key issue, how to prioritize classes of interest in a hierarchical manner. Specifically, to realize HOCD, the main idea is to design a criterion that can measure the CCP for each class of interest so that all the classes are prioritized according to their priority scores calculated by CCP. Then CCP can be further used to calculate CCP probabilities as CCP scores that will be used to prioritize classes to be detected in a hierarchical tree.

Assume that there are M classes in a hyperspectral image cube. For the first layer in the hierarchical tree, the first class with the highest CCP score is classified. Once the first class is classified, the resulting image cube is supposed to contain $M-1$ classes and BKG information. In this case, the first class information will be considered as unwanted information and then be removed from the current image cube prior to moving to the second layer to classify the second class with the second highest CCP score. The same process will be repeated over again until the $(M-1)$st layer which classifies the class with the least CCP score. A diagram of a hierarchical tree structure of HOCD can be described in Fig. 1.

Interestingly, an immediate advantage benefited from HOCD is to provide a means of dealing with imbalanced classes for classification. It is known that the small class classification has been a challenging issue in HSIC. It is closely related to the issue of how to determine the size of training samples for each of the classes. A general approach

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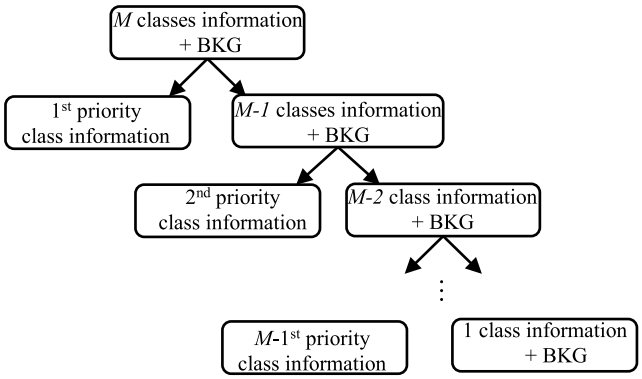


Fig. 1. Diagram of a hierarchical tree used in HOCD.

is to use sample ratio to determine the training sample size according to its class size. However, this may result in an incorrect conclusion where the classification rate of a large class may dominate the OA at the expense of the classification rate of a small class. This is a very unfortunate consequence when small classes usually represent vital and critical class information such as rare material substances, end-members, and anomalies. To mitigate this dilemma, the AA is used to consider all classes to be equally important, in which case the classification rates of small classes have the same weight as large classes. But AA still does not address the bias of training sample size of each class. Most training sample selection methods determine the number of training samples based on the sample ratio for each class. In other words, a large class will have a large size of training samples, whereas a small class will have a small size of training samples. This will result in poor performance caused by the small numbers of training samples selected for small classes. To overcome this issue, Kang et al. [30] and Kang [31] suggested a method, referred to as Kang's method, to determine the training sample size for each of the classes. Most recently, Zhong et al. [32] also developed a new concept, called class significance that can be used to allocate training size for each class. Such training sample class size allocation guarantees that small classes will have enough training samples to be used for classification. Unlike the methods mentioned above, HOCD offers a different aspect by considering a class to be classified as a desired signal, while all other data samples as BKG samples so that the signal detected in BKG is equivalent to a class of interest to be classified in BKG, which is exactly what OCD does. Since OCD performs detection, it only needs to detect the class of interest as signals and has nothing to do with the size of the class to be classified, thus the imbalanced class classification issue is resolved.

F. Contributions

Three major contributions of this article are summarized as follows.

1) In order to perform HSIC-B, a new concept of OCD derived from CEM is introduced. Also, for OCD using CEM to perform multiple-class classification for HSIC-B, HOCD is developed so that CEM can be implemented layer by layer in a hierarchical tree.
2) Since HOCD requires prioritizing classes to be classified by CEM, another new concept of CCP is further developed to calculate priority scores of classes such that classes will be processed according to their priority scores hierarchically.
3) Extensive experiments are conducted to show the effectiveness of HOCD in HSIC-B, while recently developed state of the-art (STOA) techniques that work well with HSIC-NB indeed perform poorly in HSIC-B. Most importantly, experiments also show that precision (PR) rates can be used to evaluate the degree of data BKG complexity.

This article is organized as follows. Section II reviews related studies. Section III introduces a new concept, CCP. Section IV describes key ideas of HOCD. Section V details algorithms for implementing HOCD methods. Section VI describes images to be used for experiments. Section VII presents classification measures for HSIC-NB and HSIC-B. Section VIII presents experimental results along with comparative study between proposed methods and the most recently developed state-of-the-art OCC methods. Section IX is particularly included to discuss the differences between HSIC-B and HSIC-NB. Section X draws some conclusions.

II. Related Studies

HSIC has gained significant attention across various fields evidenced by a wealth of literature, for example, target recognition, oil spill detection [33], [34], and HSIC [35], [36]. Due to utilizing hundreds of contiguous spectral bands, hyperspectral image pixel is often a mixture of multiple material signatures, both known and unknown, including elements such as shadows [37]. Such complexity challenges traditional spatial domain-based classifiers, which primarily focus on pure pixel-based analysis. This complexity is further complicated by BKG samples which are mixed with labeled data samples or entirely composed of unlabeled materials. Unfortunately, without prior knowledge many crucial material substances cannot be labeled or identified such as anomalies and end-members. On many instances, these unlabeled substances generally occur nearby or mix with class boundaries, which even present challenges for classification. Accordingly, if BKG data samples are included in HSIC, the resulting classification turns out to be very poor as shown in [19], [20], and [21]. Also, because of this reason, most HSIC techniques reported in the literature have performed classification without BKG. With BKG removed by GT, their classification performances could achieve very high classification rates. Unfortunately, when these techniques are applied to image datasets with BKG included, they indeed perform very poorly [19], [20], [21]. This evidence indicates that many HSIC techniques claimed to work well for datasets with BKG removed may actually end up performing very poorly for datasets with BKG included.

Addressing the issues arising from BKG samples, several innovational solutions have been proposed. For instance, Hao et al. [38] leveraged a fully convolutional network (FCN) for BKG segmentation, enhancing the accuracy of Glioblastoma tumor identification through the integration of multiple deep models. Similarly, Li et al. [39] introduced a nested

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spatial pyramid module to distinguish targets from the BKG by extracting multiview and multiscale features. The use of patch-wise methods is prevalent in HSIC, where larger patches provide a richer set of spectral-spatial information at the cost of increased redundancy. To counteract this, Zhao et al. [40] developed a method combining two-window discrete cosine transform and synthetic minority oversampling technique with one-versus-all ensemble classifiers, specifically addressing the dual challenges of high-dimensional data and class imbalance.

As an alternative approach, OCC has been also developed to address BKG issue in HSIC. Instead of classifying multiple classes as HSIC does, OCC focuses on identifying a single class of interest as a positive class while considering other classes as negative classes. For example, Li et al. [41] explored a positive and unlabeled learning (PUL) algorithm that requires minimal labeled positive data, while another study by Li et al. [42] proposed the positive and BKG learning with constraints (PBLC) algorithm, which directly infers the posterior probability of the positive class with a one-step model training, incorporating constraints based on maximum probability information. In addition, Zhao et al. [43] introduced a weakly supervised deep HSI one-class classifier that adapted the FCN for OCC to effectively manage distribution imbalances. Of particular interest are FSSR and FSFR using sparse representation for OCC proposed by Song et al. [21].

Despite the fact that OCC converts multiclass classification to binary classification which can deal with imbalanced class issue, it still does not address the BKG issue because BKG is neither a positive class nor a negative class. Although many DL-based methods were also developed for HSIC such as references reviewed in [2], most methods were developed for HSIC-NB. The key issue is that the complexity of BKG is extremely difficult to be characterized by training samples where many unknown material substances cannot be labeled or identified by prior knowledge and visual examination. Under this situation, computer vision-based methods are generally not applicable. So, one effective means is to treat BKG samples as unknown signal sources, thus there is no need of acquiring training samples. The problem combining OCC with the unknown BKG turns out to be an MVDR problem discussed in the introduction where the positive class is considered as signals arriving from the desired direction, $\mathbf{d}$, while the negative classes and BKG can be treated as signals arriving from undesired directions. Minimizing the energy resulting from signals coming all directions other than from $\mathbf{d}$ is equivalent to minimizing the least squares error of misclassifying the data samples in negative classes and BKG. The first work to realize this concept was envisioned in [44] where an ICEM was specifically developed for a single class classification HSIC. Then ICEM was further extended to ITCIMC for HSIC [45] where TICMC is actually the TCIMF [25] specifically used for multiclass classification. Since CEM and TCIMF are linear filters, they may not effectively deal with data samples on class boundaries or mixed with BKG when they are used for HSIC. To mitigate this dilemma, CEM and TCIMF were further extended to KCEM [46] and KTCIMF [47], both of which will be used to design HOCD.

III. Class Classification Priority (CCP)

The concept of CCP can be traced back to CEM which is used to design HOCD. To see this, CEM is first reviewed and then the result of CEM is further used to design CCP.

A. CEM

Suppose that a hyperspectral image is represented by a collection of image pixel vectors, denoted by $\{\mathbf{r}_i\}_{i=1}^N$ with $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ being an L-dimensional pixel vector and L being the total number of spectral bands. More specifically, let the original HSI sample space be Ω_{data}, where $\Omega_{\text{data}} = \{\mathbf{r}_n\}_{n=1}^N$ and $\mathbf{r}_n \in \Re^{L\times 1}$ be an nth data sample with L bands. Let $\delta(\mathbf{r}_n)$ be a linear filter specified by a weighting vector $\mathbf{w} = (w_1, w_2, \ldots, w_L)^T$ as

$$\delta(\mathbf{r}_n) = \mathbf{w}^T\mathbf{r}_n. \tag{1}$$

Then its averaged output energy over the entire image scene is represented by

$$\frac{1}{N}\sum_{n=1}^{N}\delta^2(\mathbf{r}_n) = \mathbf{w}^T\mathbf{R}\mathbf{w} \tag{2}$$

where $\mathbf{R} = (1/N)\sum_{n=1}^N \mathbf{r}_n\mathbf{r}_n^T \in \Re^{L\times L}$ is the sample autocorrelation matrix, N is the total number of data samples, and L is the number of spectral bands.

Now, assume that $\mathbf{d}$ is the desired target signature used to constrain the linear filter (1). We can formulate the following constrained optimization problem as follows:

$$\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\}, \quad \text{s.t. } \mathbf{d}^T\mathbf{w} = 1. \tag{3}$$

Solution to (3) is given by

$$\mathbf{w}^{\text{CEM}} = \mathbf{R}^{-1}\mathbf{d}\left(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}\right)^{-1}. \tag{4}$$

Substituting (4) into (3) yields the CEM detector, $\delta^{\text{CEM}}(\mathbf{r})$ developed in [22] and given by

$$\delta^{\text{CEM}}(\mathbf{r}) = \left(\mathbf{w}^{\text{CEM}}\right)^T\mathbf{r} = \frac{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}}. \tag{5}$$

Let the resulting whitened data space be denoted by $\bar{\mathbf{X}}$ and the whitened data sample be denoted by

$$\bar{\mathbf{r}} = \mathbf{R}^{-1/2}\mathbf{r}. \tag{6}$$

The linear filter (1) is then operated on $\bar{\mathbf{X}}$ to yield

$$\delta_{\mathbf{w}}(\bar{\mathbf{r}}) = \left(\bar{\mathbf{w}}^{\text{CEM}}\right)^T\bar{\mathbf{r}} \tag{7}$$

where (4) becomes $\bar{\mathbf{w}}^{\text{CEM}} = \bar{\mathbf{d}}^T$ with $\bar{\mathbf{d}} = \mathbf{R}^{-1/2}\mathbf{d}$. Then the SNR of (7) in $\bar{\mathbf{X}}$ can be calculated by

$$\begin{aligned}\text{SNR}(\delta^{\text{CEM}}(\bar{\mathbf{r}})) &= \frac{\left(\bar{\mathbf{w}}^{\text{CEM}}\right)^T\bar{\mathbf{r}}\left(\left(\bar{\mathbf{w}}^{\text{CEM}}\right)^T\bar{\mathbf{r}}\right)^T}{\left[E\left(\left(\mathbf{w}^{\text{CEM}}\right)^T\bar{\mathbf{n}}\right)^2\right]} \\ &= \frac{\left(\bar{\mathbf{d}}^T\bar{\mathbf{r}}\right)^2}{\left[E\left(\bar{\mathbf{d}}^T\bar{\mathbf{n}}\right)^2\right]}.\end{aligned} \tag{8}$$

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By Schwarz's inequality, (8) becomes

$$\mathrm{SNR}(\delta^{\mathrm{CEM}}(\bar{\mathbf{r}})) \leq \frac{\left(\bar{\mathbf{d}}^T\bar{\mathbf{d}}\right)^2}{\left[E\left(\bar{\mathbf{d}}^T\bar{\mathbf{n}}\right)^2\right]} \tag{9}$$

where the equality holds if and only if $\bar{\mathbf{r}}_n^* = \kappa\bar{\mathbf{d}}$ for any constant κ. Equation (9) implies that the max SNR of CEM can be obtained by (9) as

$$\mathrm{SNR}(\delta_{\mathbf{w}^*}(\mathbf{r})) = \kappa\bar{\mathbf{d}}^T\bar{\mathbf{d}} = \kappa\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}. \tag{10}$$

By letting $\kappa = 1$, (10) is reduced to the SNR of CEM in (7)

$$\mathrm{SNR}(\delta^{\mathrm{CEM}}(\mathbf{r})) = \mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}. \tag{11}$$

Different from CEM, TCIMF is also an OCD method that detects the desired target and eliminates the undesired signatures at the same time while CEM only focuses on the desired target. TCIMF was developed in [25] based on the concept from the combination of CEM and OSP by applying the constraint vector to a desired target $\mathbf{d}$ and undesired targets $\mathbf{U}$.

B. LCMV

In CEM, the MVDR filter used a single constraint to constrain a desired target signature $\mathbf{d}$. In [48], Frost extended CEM to LCMV filter, which allows multiple linear constraints to adjust an array of sensors in real time to respond to a set of desired signals specified by $[\mathbf{D}]$ while discriminating against noise. Using multiple constraints on weights of $\mathbf{D}$ in the array, LCMV filter can be shown to detect multiple targets with no need of prior knowledge of undesired signals or noise statistics. The resulting LCMV filter is called LCMV target detector.

Assume that $\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_M$ are M specific target signatures of interest, which can be either provided by a priori knowledge or obtained by training samples. Suppose that $\{\mathbf{r}_i\}_{i=1}^N$ is the set of data sample vectors in a hyperspectral image where $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ is the ith L-dimensional data sample vector and L is the total number of spectral bands.

Now, we interpret each of $\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_M$ as a desired signal arrival direction in adaptive beamforming and let $\mathbf{D}^M = [\mathbf{d}_1\,\mathbf{d}_2\cdots\mathbf{d}_M]$ be an $L \times M$ target signature matrix where L is the total number of band. Assume that an finite impulse response (FIR) linear filter is specified by L filter coefficients $\{w_1, w_2, \ldots, w_L\}$, denoted by an L-dimensional vector $\mathbf{w} = (w_1, w_2, \ldots, w_L)^T$. The LCMV problem can be then reformulated as the following M-target constrained optimization problem

$$\min_{\mathbf{w}}\left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\}, \quad \text{s.t. } \left(\mathbf{D}^M\right)^T\mathbf{w} = \mathbf{c}^M \tag{12}$$

where $\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\}$ in (12) was referred to as minimum variance, $\mathbf{c}^M = (c_1, c_2, \ldots, c_M)^T$ is an M-dimensional constraint vector for a general purpose, and $\mathbf{R} = (1/N)\sum_{i=1}^N \mathbf{r}_i\mathbf{r}_i^T$ is the sample correlation matrix of size $L \times L$. It should be noted that each of M constraints, $c_1, c_2, \ldots, c_M$, in (12) is used to impose on a particular target, i.e., the mth constraint, c_m is imposed on the mth target, $\mathbf{d}_m$, and chosen to be any arbitrary M-dimensional vector. For example, in adaptive beamforming of array signal processing $\mathbf{c}^M$ is only used to lock signal arrivals in M desired directions. For simplicity, we can choose $\mathbf{c}^M$ to be the M-dimensional unity vector with ones in all the M components, $\mathbf{1}^M = (1, \ldots, \underbrace{1}_{j}, \ldots, \underbrace{1}_{M})^T$. In this case, the optimal solution to (12) with $\mathbf{c}^M$ replaced by $\mathbf{1}^M$ is solved by

$$\mathbf{w}^* = \mathbf{R}^{-1}\mathbf{D}^M\left(\left(\mathbf{D}^M\right)^T\mathbf{R}^{-1}\mathbf{D}^M\right)^{-1}\mathbf{1}^M \tag{13}$$

where $(\mathbf{D}^M)^T\mathbf{R}^{-1}\mathbf{D}^M$ is an $M \times M$ matrix.

Then LCMV is specified by

$$\delta^{\mathrm{LCMV}}(\mathbf{r}) = \left(\mathbf{w}^*\right)^T\mathbf{r} \tag{14}$$

which performs multitarget detection with M target signature matrix $\mathbf{D}^M$ constrained by $\mathbf{1}^M$ and BKG suppressed by $\mathbf{R}^{-1}$.

Now, substituting (13) into (12) yields

$$\left(\mathbf{w}^*\right)^T\mathbf{R}\mathbf{w}^* = \left(\mathbf{1}^M\right)^T\left(\left(\mathbf{D}^M\right)^T\mathbf{R}^{-1}\mathbf{D}^M\right)^{-1}\mathbf{1}^M \tag{15}$$

which is the minimum variance resulting from LCMV in (14).

C. TCIMF

In many practical applications, there are M desired targets specified by $\mathbf{D}^M$ and q undesired target signatures, denoted by $\mathbf{U} = [\mathbf{u}_1\mathbf{u}_2\ldots\mathbf{u}_q]^T$. Then LCMV can be further extended to TCIMF developed in [25] to find an optimal $\mathbf{w}$ to minimize the averaged output energy by solving the following formula:

$$\min_{\mathbf{w}}\left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\}, \quad \text{s.t } \left[\mathbf{D}^M\mathbf{U}\right]^T\mathbf{w} = \begin{bmatrix}\mathbf{1}_{M\times 1}\\ 0_{q\times 1}\end{bmatrix} \tag{16}$$

where $\mathbf{1}_{M\times 1}$ is used to constrain the desired target matrix $\mathbf{D}^M$ as CEM does, and $\mathbf{0}_{q\times 1}$ is used to suppress the undesired targets $\mathbf{U}$. The optimal solution to (16) is given by

$$\mathbf{w}^{\mathrm{TCIMF}} = \mathbf{R}^{-1}\left[\mathbf{D}^M\mathbf{U}\right]\left(\left[\mathbf{D}^M\mathbf{U}\right]^T\mathbf{R}^{-1}\left[\mathbf{D}^M\mathbf{U}\right]\right)^{-1}\begin{bmatrix}1_{M\times 1}\\ 0_{q\times 1}\end{bmatrix}. \tag{17}$$

Then, TCIMF is given by

$$\delta^{\mathrm{TCIMF}}(\mathbf{r}) = \left(\mathbf{w}^{\mathrm{TCIMF}}\right)^T\mathbf{r}. \tag{18}$$

Noted that the constraint vector

$$\begin{bmatrix}1_{M\times 1}\\ 0_{q\times 1}\end{bmatrix}$$

in (17) could also be replaced by a constraint matrix $\mathbf{C}$ which is specified by

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 & \vdots \\ \vdots & 0 & \ddots & \ddots & \vdots & 0 \\ 0 & \vdots & \ddots & 1 & 0 & \vdots \\ \vdots & 0 & \cdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & \cdots & 0 & 1 \end{bmatrix}_{M\times M}. \tag{19}$$

If we incorporate $\mathbf{C}$ in (19) into TCIMF, then the output becomes an M-dimensional column vector including not only the detection result of $\mathbf{D}^M$ but also the annihilation results of $\mathbf{U}$.

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D. KCEM/KTCIMF

Both CEM and TCIMF are linear filters which are hard to deal with nonlinear separability problem. Recently, kernel method has been widely used due to its ability to deal with nonlinear problem, which use a nonlinear function that maps the original linearly inseparable data to a high dimensional space to make data separable. Assume the input data $\mathbf{x}$ are mapped into a feature space F with a nonlinear function Φ: $\Re^N \rightarrow F$ by $\mathbf{x} \rightarrow \Phi(\mathbf{x})$, and the kernel-based CEM [46] is designed by

$$\begin{aligned}\delta^{\mathrm{KCEM}}(\mathbf{r}) &= \left(\mathbf{w}^{\mathrm{KCEM}}\right)^T \Phi(\mathbf{r}) \\ &= \Phi(\mathbf{d})^T \mathbf{R}_\Phi^{-1} \Phi(\mathbf{r}) \left(\Phi(\mathbf{d})^T \mathbf{R}_\Phi^{-1} \Phi(\mathbf{d})\right)^{-1}. \end{aligned} \tag{20}$$

Then, the kernel-based TCIMF [47] is designed by

$$\begin{aligned}\delta^{\mathrm{KTCIMF}}(\mathbf{r}) &= \left(\mathbf{w}^{\mathrm{KTCIMF}}\right)^T \Phi(\mathbf{r}) \\ &= \frac{\Phi\left(\left[\mathbf{D}^M \mathbf{U}\right]^T\right) \mathbf{R}_\Phi^{-1} \Phi(\mathbf{r})}{\Phi\left(\left[\mathbf{D}^M \mathbf{U}\right]^T\right) \mathbf{R}_\Phi^{-1} \Phi\left(\left[\mathbf{D}^M \mathbf{U}\right]\right)} \mathbf{C}_\Phi \end{aligned} \tag{21}$$

where the symbol with Φ indicates that they are in the feature space F. Since (20) and (21) are impossible to solve due to the infinite dimensionality in feature space. As a consequence, the RBF kernel function is used to calculate (20) and (21). Using the derivation in [46], the KCEM can be derived as

$$\delta^{\mathrm{KCEM}}(\mathbf{r}) = \frac{\mathbf{k}_\mathbf{d}^T \mathbf{K}^{-2} \mathbf{k}_\mathbf{r}}{\mathbf{k}_\mathbf{d}^T \mathbf{K}^{-2} \mathbf{k}_\mathbf{d}} \tag{22}$$

where $\mathbf{k}_\mathbf{r} = (K(\mathbf{r}, \mathbf{r}_1), K(\mathbf{r}, \mathbf{r}_2), \ldots, K(\mathbf{r}, \mathbf{r}_N))^T$ is a column vector that contains the similarities of $\mathbf{r}$and all data samples; $\mathbf{k}_\mathbf{d} = (K(\mathbf{d}, \mathbf{r}_1), K(\mathbf{d}, \mathbf{r}_2), \ldots, K(\mathbf{d}, \mathbf{r}_N))^T$ is a column vector that includes all the kernels between $\mathbf{d}$ and $\mathbf{X}$ with the RBF kernel $K(\mathbf{r}_n, \mathbf{r}_j) = \exp(-\|\mathbf{r}_n - \mathbf{r}_j\|^2/2\sigma^2)$. The output of KTCIMF constrained by (19) is given by

$$\delta^{\mathrm{KTCIMF}}(\mathbf{r}) = \frac{\mathbf{k}_{[\mathbf{D}^M \mathbf{U}]}^T \mathbf{K}^{-2} \mathbf{k}_\mathbf{r}}{\mathbf{k}_{[\mathbf{D}^M \mathbf{U}]}^T \mathbf{K}^{-2} \mathbf{k}_{[\mathbf{D}^M \mathbf{U}]}} \mathbf{C}. \tag{23}$$

However, manipulating the kernel matrix $\mathbf{K}$ is very expensive in terms of computational times and memories. In this case, low-rank approximation such as Nyström method [49], [50], [51] can be used to relax the computation cost of (22) and (23).

E. Sample Correlation-Based CCP

The idea of CCP is actually derived from (11) which is the SNR of CEM.

Assume that there are M classes of interest, $\{C_m\}_{m=1}^M$ and μ_i is the sample mean of the mth class, C_m. We can use the SNR given in (11) to define a CCP criterion as

$$\rho_R^{\mathrm{CCP}}(C_m) = \boldsymbol{\mu}_m^T \mathbf{R}^{-1} \boldsymbol{\mu}_m \tag{24}$$

which is indeed the spectral angle between $\boldsymbol{\mu}_m$ and $\mathbf{R}^{-1}\boldsymbol{\mu}_m$. It should be noted that we can rewrite (24) as $\boldsymbol{\mu}_m^T \mathbf{R}^{-1} \boldsymbol{\mu}_m = (\mathbf{R}^{-1/2}\boldsymbol{\mu}_m)^T (\mathbf{R}^{-1/2}\boldsymbol{\mu}_m) = \bar{\boldsymbol{\mu}}_m^T \bar{\boldsymbol{\mu}}_m$ with $\bar{\boldsymbol{\mu}}_m = \mathbf{R}^{-1/2}\boldsymbol{\mu}_m$ defined in [10, eq. (43)] and [11, eq. (45)]. In this case, $\rho_R^{\mathrm{CCP}}(C_m) = \bar{\boldsymbol{\mu}}_m^T \bar{\boldsymbol{\mu}}_m$ in (24) actually represents the energy of the mth class, C_m in the $\mathbf{R}^{-1/2}$-whitened space, which is completely determined by the class mean and has nothing to do with class size. As a result, the greater the value of (24) is, the higher the priority of C_m is regardless of whether its class size is small or large. Accordingly, using (24) to measure the priorities of classes is well-justified.

More interestingly, if we further assume that $\mathbf{R}$ is zero-mean and independent with $\mathbf{R} = \sigma^2 \mathbf{I}$ where σ^2 is the variance and $\mathbf{I}$ is the identity matrix, then (24) becomes $\boldsymbol{\mu}_m^T \mathbf{R}^{-1} \boldsymbol{\mu}_m = \|\mathbf{R}^{-1/2}\boldsymbol{\mu}_m\|^2 = \|\|\boldsymbol{\mu}_m\|^2/\sigma^2$ which turns out to be the standard SNR in communications and signal processing. So, the higher is SNR, the higher is the priority, $p^{\mathrm{CCP}}(C_m)$.

Using (24) we can calculate the CCP probability of the mth class, C_m as follows:

$$p_m^{\mathrm{CCP}} = p^{\mathrm{CCP}}(C_m) = \frac{\rho_R^{\mathrm{CCP}}(C_m)}{\sum_{i=1}^M \rho_R^{\mathrm{CCP}}(C_i)} \tag{25}$$

which can be used to prioritize C_i according to the magnitude of its $\mathbf{R}$-decorrelated class mean, $\|\mathbf{R}^{-1/2}\boldsymbol{\mu}_m\|$. This implies that the higher the value of $p^{\mathrm{CCP}}(C_m)$ is, the better the C_m is to be classified correctly.

IV. Ideas of HOCD

Although LCMV and TCIMF can perform multiclass classification via a desired target matrix $\mathbf{D}^M$ in a single shot, after all, they are only detection but not classification techniques. As a result, they cannot perform as well as HSIC which is specifically designed for multiclass classification. This is because when LCMV/TCIMF are used for multiclass classification, they actually perform multisignal detection and do not use training samples as HSIC does to classify data samples. In this case, the class correlation among data samples provided by training samples cannot be used to separate detected classes. This is exactly why we introduce HOCD with CCP to replace OCD so that each class can be detected by OCD in a hierarchical tree as a separate signal, while CCP can be used to remove class similarity to enhance class separability.

Three main key ideas to implement HOCD are briefly described in the following three steps.

Step 1: Computation of CCP Scores for Class Prioritization Calculate the mean of each class of $\{C_m\}_{m=1}^M$ by GT samples (i.e., labeled data samples) and then use these means to compute a CCP score for each of M classes via (25), i.e., $\{p_m\}_{m=1}^M$. Let $\{\boldsymbol{\mu}_m\}_{m=1}^M$ be class means to be arranged in a descending order according to $\{p_m\}_{m=1}^M$.

Step 2: Inner loop with an iterative process indexed by k to classify a given prioritized class obtained by Step 1 in a hierarchical layer.

At the mth layer in a hierarchical tree, randomly select a set of training samples to calculate the mth class mean $\boldsymbol{\mu}_m$ which will be used to calculate the desirable signature $\mathbf{d}$ for two CEM-based OCD methods, KCEM or KTCIMF.

1) *KCEM:* Implement KCEM to produce a probability map $\mathrm{SMap}_{\mathrm{KCEM},m}^{(k)}$. Then, apply Otsu's method [52] to the soft probability map of C_m, $\mathrm{SMap}_{\mathrm{KCEM},m}^{(k)}$, to produce a binary map $\mathrm{BMap}_m^{(k)}$ to check if it satisfies a stopping rule for inner loop. If not, feed the $\mathrm{SMap}_{\mathrm{KCEM},m}^{(k)}$ back and add

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to the current data cube to generate a new data cube for next iteration $k+1$. Otherwise, go to outer loop.

2) *KTCIMF:* Implement KTCIMF to produce the soft probability map $\text{SMap}^{(k)}_{\text{KTCIMF},m}$. Then, apply Otsu's method to binarize $\text{SMap}^{(k)}_{\text{KTCIMF},m}$ to produce a binary map $\text{BMap}^{(k)}_{m}$ to see if it satisfies a stopping rule for inner loop. If not, feed $\text{SMap}^{(k)}_{\text{KTCIMF},m}$ back and add to the current data cube to generate a new data cube for next iteration $k+1$. Otherwise, go to outer loop.

Step 3: Outer loop indexed by $1 \le m \le M$ for C_m

Once a class is classified in Step 2, two operations are carried out.

1) The first operation is to remove the means that are similar to the mth class mean, $\boldsymbol{\mu}_m$ from previous classified classes, i.e., $\{\hat{C}_i\}_{i=1}^{m-1}$ in terms of spectral similarity. Let $\mathbf{U}_{m-1} = [\boldsymbol{\mu}_1 \boldsymbol{\mu}_2 \cdots \boldsymbol{\mu}_{n_{m-1}}]$ be the matrix made up of all the previous classified class means where n_m is the number of dissimilar classified class means kept in $\mathbf{U}_{m-1}$ with $1 \le n_{m-1} \le m-1$. Then an undesired signature projection matrix $P^{\perp}_{\mathbf{U}_{m-1}}$ defined by

$$P^{\perp}_{\mathbf{U}_{m-1}} = \mathbf{I} - \mathbf{U}_{m-1}\left(\mathbf{U}^T_{m-1}\mathbf{U}_{m-1}\right)^{-1}\mathbf{U}^T_{m-1} \tag{26}$$

to remove classification information obtained from previous layers in HOCD so as to improve the classification rate of subsequent layers. However, there is an issue arising from (26). If the mean, $\boldsymbol{\mu}_m$ of C_m is very similar to the means of previously removed classified classes in $\mathbf{U}_{m-1}$, then the most class information of C_m will be removed by (26). This will result in significant reduction of classification rate of C_m. That is, if the similarity between $\boldsymbol{\mu}_m$ and μ in $\mathbf{U}_{m-1}$ is too high, the crucial class information will be removed by $P^{\perp}_{\mathbf{U}_{m-1}}$ in (26). To avoid this situation, spectral information divergence (SID) developed in [53] and [54] is then used to calculate class similarity between the mth class mean and the class means in $\mathbf{U}_{m-1}$ by

$$\left\{\text{SID}\left(\boldsymbol{\mu}_m, \boldsymbol{\mu}\right)\right\}_{\boldsymbol{\mu}\in\mathbf{U}_{m-1}} \tag{27}$$

determine which means must be removed from $\mathbf{U}_{m-1}$. If $\text{SID}(\boldsymbol{\mu}_m, \boldsymbol{\mu}) < \tau$ in (27) for some μ in $\mathbf{U}_{m-1}$ with a prescribed error threshold τ, then μ will be removed from $\mathbf{U}_{m-1}$ prior to implementing (25). In this case, $\mathbf{U}_{m-1}$ will be replaced by $\mathbf{U}_m = \mathbf{U}_{m-1} - \{\boldsymbol{\mu}\}$. This process runs through all the $\boldsymbol{\mu}$ in $\mathbf{U}_{m-1}$ and updates $\mathbf{U}_{m-1}$.

2) The second operation is to suppress BKG effect on classification rate of the mth class, C_m via inverting the correlation matrix $\mathbf{R}$.

According to (7), CEM makes use of $\mathbf{R}^{-1}$ to suppress BKG impact on classification. So, when KCEM and KTCIMF are used to classify the mth class, C_m, their use of the correlation matrix $\mathbf{R}$ should be formed by those data samples that need to be suppressed. Let $\boldsymbol{\Omega}_m$ be the set of data samples formed by

$$\boldsymbol{\Omega}_m = \hat{C}_1 \cup \cdots \cup \hat{C}_{n_m} \cup C_m. \tag{28}$$

Then $\mathbf{R}$ is replaced by $\mathbf{R}_{m-1}$ given by

$$\mathbf{R}_{m-1} = \frac{1}{|\boldsymbol{\Omega} - \boldsymbol{\Omega}_m|}\sum_{\mathbf{r}\in\boldsymbol{\Omega}-\boldsymbol{\Omega}_m}\mathbf{r}\mathbf{r}^T \tag{29}$$

where $\boldsymbol{\Omega} = \{\mathbf{r}_n\}_{n=1}^N$ is the set of total data samples.

V. Algorithm for HOCD

Using the above three key steps, algorithms, called HOCD-IRTS-KCEM and HOCD-IRTS-KTCIMF can be designed and developed for HSIC-B. The details of algorithms that implement HOCD-IRTS-KCEM/KTCIMF step by step are described in the following.

Fig. 2 depicts a generic block diagram of implementing HOCD-IRTS-KCEM/KTCIMF. It describes three major steps implemented in Algorithm 1 where the block of step 1 calculates CCP scores as an initial condition; the block of step 2 details the iterative process carried out in the inner loop for each class; and the block of step 3 is an outer loop to process class by class hierarchically.

Upon implementing Step 3 in Algorithm 1, two important operations, $P^{\perp}_{\mathbf{U}_{m-1}}$ in (26) and $\mathbf{R}_{m-1}$ in (29) are designed to remove overlapped classification information from previous classified classes and suppress unknown BKG respectively. On some occasions, these two operations may not be all needed. To deal with this dilemma, two special cases of implementing Algorithm 1 can be developed in the following versions.

1) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM/KTCIMF which utilizes $\mathbf{R}_{m-1}$ with $\hat{\mathbf{X}}_m = \mathbf{X}$ without using $P^{\perp}_{\mathbf{U}_{m-1}}$ for $1 \le m \le M-1$.
2) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM/KTCIMF which only utilizes $P^{\perp}_{\mathbf{U}_{m-1}}$ but still uses the global sample correlation matrix $\mathbf{R}$ in (2).

To be consistent with the above notations, the complete version of HOCD implemented in Algorithm 1 is then referred to as $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM/KTCIMF which utilizes both $\mathbf{R}_{m-1}$ and $P^{\perp}_{\mathbf{U}_{m-1}}$ with $\hat{\mathbf{X}}_m = P^{\perp}_{\mathbf{U}_{m-1}}\mathbf{X}$.

VI. Images to be Used for Experiments

In this section, four popular real hyperspectral images were used for experiments to conduct comparative analyses.

A. Purdue Indiana Indian Pines

The first one is the Purdue Indiana Indian Pines test site with an aerial view shown in Fig. 3(a) along with a USGS quadrangle map of the test site in Fig. 3(a) and its GT of 17 class maps in Fig. 3(c) and their class labels in Fig. 3(d). In general, BKG pixels can be referred to as uninteresting data samples or unlabeled data samples which cannot be identified by GT. Obviously, this scene was particularly collected with an intention of classifying 16 classes of crops. In this case, the BKG is considered to be those made up of all other noncrop data samples including radio tower, roads, highways, and railroads, according to USGS quadrangle map shown in Fig. 3(a). However, if the Purdue data is used for other applications, such as land use/cover and military applications,

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Algorithm 1 HOCD-IRTS-KCEM/KTCIMF

1) Initial condition: Let $\mathbf{X} = \{\text{HSI}\}$ and $\{C_m\}_{m=1}^{M}$ be the prioritized classes by $\{p_m\}_{m=1}^{M}$ where M is number of classes. Set $m = 1$, then $\hat{\mathbf{X}}_1^{(0)} = \mathbf{X}$.
2) Outer Loop:
   - If $m = 1$, go to step 3. Otherwise, let $\mathbf{U}_{m-1} = \left[\boldsymbol{\mu}_{j_1}\boldsymbol{\mu}_{j_2}\cdots\boldsymbol{\mu}_{j_{m-1}}\right]$ and update $\mathbf{R}_m$ by (29).
   - Check if $\text{SID}(\boldsymbol{\mu}_m, \boldsymbol{\mu}) < \tau$ for all $\boldsymbol{\mu} \in \mathbf{U}_{m-1}$. If yes, remove μ from $\mathbf{U}_{m-1}$ and the class corresponding to μ from $\mathbf{R}_m$ to update$\mathbf{U}_{m-1}$ and $\mathbf{R}_m$ by (29).
   - Apply $P^{\perp}_{\mathbf{U}_{m-1}}$ in (26) to the original data matrix $\mathbf{X}$ to produce $\hat{\mathbf{X}}_m = P^{\perp}_{\mathbf{U}_{m-1}}\mathbf{X}$.
   - Check if $m = M$. If yes, then go to step 6. Otherwise, go to step 3.
3) Inner Loop:
   - At the kth iteration, randomly select a new set of training samples of C_m, $\text{S}^{(k)}_{\text{Training},m}$.
   - There are two types of classifiers, KCEM and KTCIMF, can be applied as follows.
     a) KCEM:
     - Implement KCEM on the data matrix $\hat{\mathbf{X}}_m^{(k)}$ using $\mathbf{R}_m$ and $\text{S}^{(k)}_{\text{Training},m}$ to classify C_m to produce a soft probability map $\text{SMap}^{(0)}_{\text{KCEM},m}$.
     - Apply Otsu's method to $\text{SMap}^{(0)}_{\text{KCEM},m}$ and produce $\hat{C}_m$ and a binary map $\text{BMap}^{(0)}_{\text{KCEM},m}$. Specifically, the value of pixel at (x, y) in the binary map is defined as follows:

$$\text{BMap}^{(0)}_{\text{KCEM},m}(x,y) = \begin{cases} 1; & \text{if pixel at } (x,y) \in \text{ class } C_m \\ 0; & \text{otherwise} \end{cases} \tag{30}$$

     b) KTCIMF:
     - Implement KTCIMF on the data matrix $\hat{\mathbf{X}}_m^{(k)}$ using $\mathbf{R}_m$ and $\text{S}^{(k)}_{\text{Training},m}$ to classify C_m to produce M soft probability maps $\left\{\text{SMap}^{(0)}_{\text{KTCIMF},j}\right\}_{j=1}^{M}$. The soft probability map of C_m is defined as $\text{SMap}^{(0)}_{\text{KTCIMF},m}$.
   - Apply Otsu's method to binarize $\text{SMap}^{(0)}_{\text{KTCIMF},m}$ and produce $\hat{C}_m$ and a binary map $\text{BMap}^{(0)}_{m}$. Specifically, the value of pixel at (x, y) in the binary map is defined as follows:

$$\text{BMap}^{(0)}_{\text{KTCIMF},m}(x,y) = \begin{cases} 1; & \text{if pixel at } (x,y) \in \text{ class } C_m \\ 0; & \text{otherwise} \end{cases} \tag{31}$$

4) a) KCEM:
   - At the k-th iteration, apply Otsu to $\text{SMap}^{(k)}_{\text{KCEM},j}$ and produce a binary map $\text{BMap}^{(k)}_{\text{KCEM},j}$, i.e.,

$$\text{BMap}^{(k)}_{\text{KCEM},j}(x,y) = \begin{cases} 1; & \text{if pixel at } (x,y) \in \text{ class } C_j \\ 0; & \text{otherwise} \end{cases} \tag{32}$$

Algorithm 1 *(Continued.)* HOCD-IRTS-KCEM/KTCIMF

   b) KTCIMF:
   - At the k-th iteration, apply Otsu's method to binarize $\text{SMap}^{(k)}_{\text{KTCIMF},j}$ and produce a binary map $\text{BMap}^{(k)}_{\text{KTCIMF},j}$,i.e.,

$$\text{BMap}^{(k)}_{\text{KTCIMF},j}(x,y) = \begin{cases} 1; & \text{if pixel at } (x,y) \in \text{ class } C_j \\ 0; & \text{otherwise} \end{cases} \tag{33}$$

5) a) KCEM:
   - Generate a new data cube

$$\hat{\mathbf{X}}_m^{(k)} = \hat{\mathbf{X}}_m^{(k-1)} \cup \text{SMap}^{(k)}_{\text{KCEM},m}. \tag{34}$$

   b) KTCIMF:
   - Generate a new data cube

$$\hat{\mathbf{X}}_m^{(k)} = \hat{\mathbf{X}}_m^{(k-1)} \cup \left\{\text{SMap}^{(k)}_{\text{KTCIMF},j}\right\}_{j=1}^{M}. \tag{35}$$

   - If $k = 1$ go back to step 3. Otherwise, check if $\text{BMap}^{(k)}_{\text{KCEM/KTCIMF},m}$ satisfies a stopping rule,

$$\text{TI}^{(m)} = \frac{\left|\text{BMap}^{(k)}_{\text{KCEM/KTCIMF},m} \cap \text{BMap}^{(k-1)}_{\text{KCEM/KTCIMF},m}\right|}{\left|\text{BMap}^{(k)}_{\text{KCEM/KTCIMF},m} \cup \text{BMap}^{(k-1)}_{\text{KCEM/KTCIMF},m}\right|} > \varepsilon \tag{36}$$

     , where $\text{TI}^{(m)}$ is Tanimoto index (TI) [55] for the mth class, C_m If yes, let $m \leftarrow m + 1$ and go to step 2. Otherwise, let $k \leftarrow k + 1$ and go to step 3.

6) Output: The algorithm is terminated and the $\left\{\text{BMap}^{(k)}_{\text{KCEM/KTCIMF},m}\right\}_{m=1}^{M}$ is the final classification maps.

crop classification will be no longer of interest. Instead, the radio tower, roads, highways, and railroads will be of main interests. This indicates that without knowing specific applications, BKG should be included for classification.

Table I tabulates the number of data samples in each class where there are four small classes with less than 100 samples, classes 9, 7, 1, 16, and three classes with more than 1000 samples, classes, 2, 11, 14. So, this scene clearly has an imbalanced class issue in classification. It is an airborne visible/infrared imaging spectrometer (AVIRIS) image scene with 20-m spatial resolution and has size of 145 × 145 × 200 pixel vectors with water absorption bands (bands 104–108 and 150–163, 200). So, a total of 200 bands were used for experiments. It should be noted that in many reports, 200 bands were used by excluding water absorption bands such as [22].

B. Salinas

A second image dataset is Salinas in Fig. 4(a) which is also an AVIRIS scene. It was collected over Salinas Valley, California, with a spatial resolution of 3.7 m/pixel with spectral resolution of 10 nm. It has a size of

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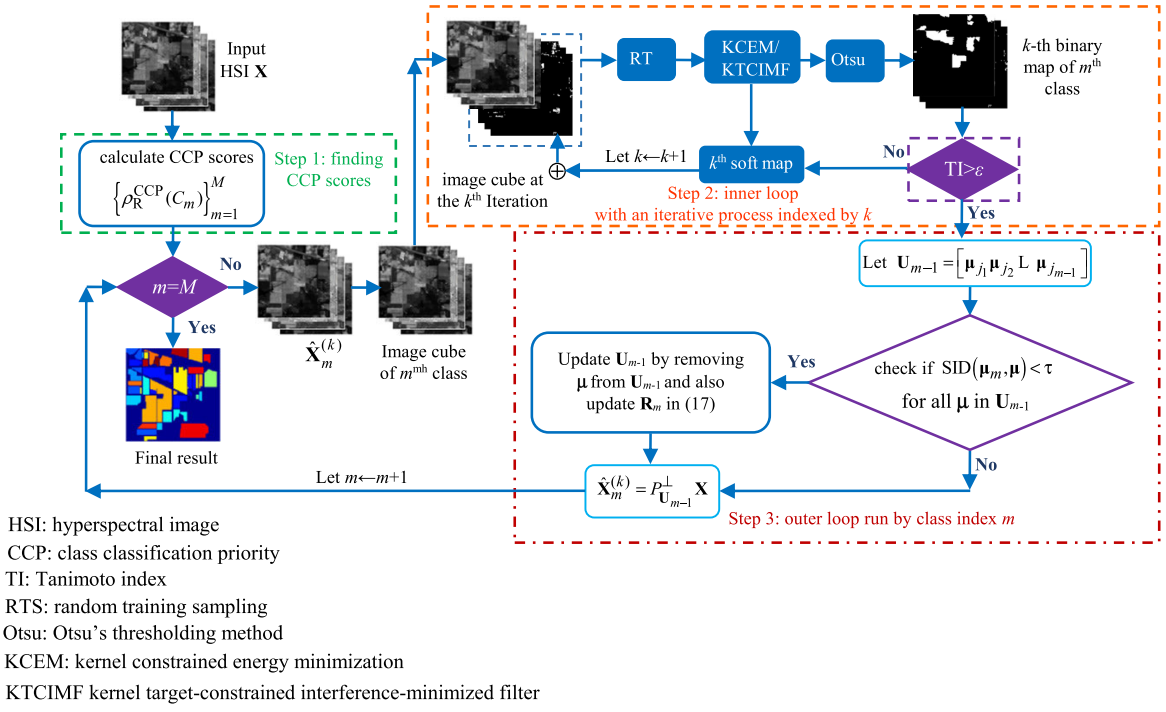


Fig. 2. General flowchart of implementing HOCD-IRTS-KCEM/KTCIMF.

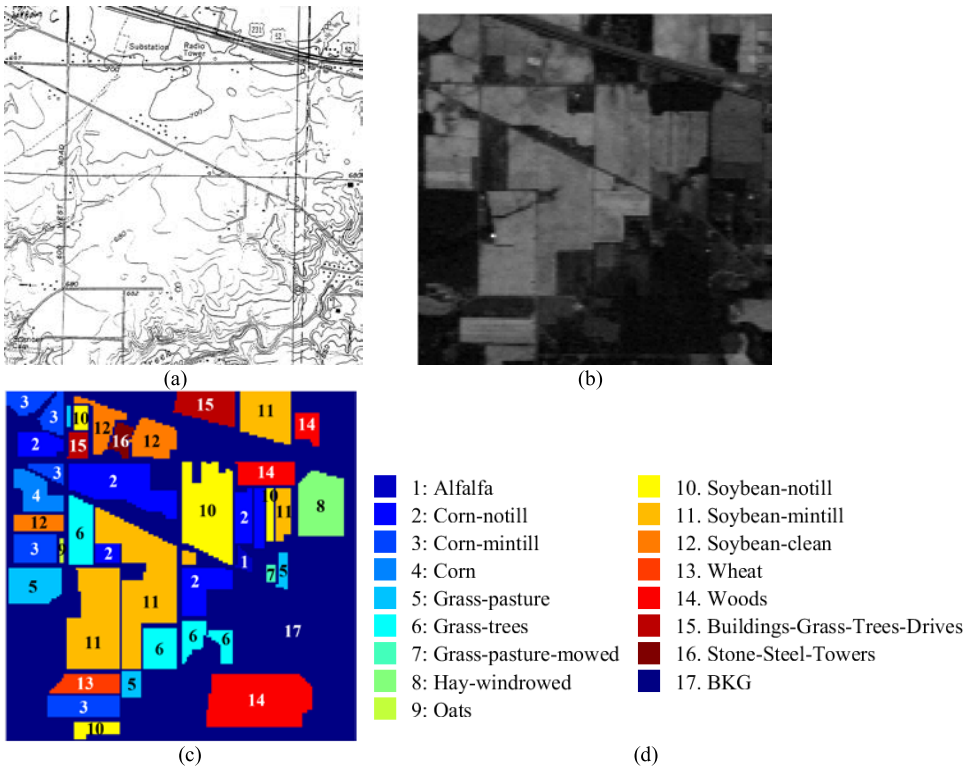


Fig. 3. Purdue's data and its GT with 16 classes. (a) USGS quadrangle map of the test site. (b) Band 186 (2162.56 nm). (c) Ground-truth map. (d) Classes by colors.

512 × 217 × 224 with 20 water absorption bands, 108–112, 154–167, and 224. So, a total of 204 bands were used for experiments. Fig. 4(b) and (c) shows the color composite of the Salinas image along with the corresponding GT class labels. Unlike the Purdue data Salinas scene does not have an issue in imbalanced classes. According to the GT tabulated in Table II, the smallest class is class 13 with 916 data samples.

C. University of Pavia

A third hyperspectral image data was the University of Pavia image shown in Fig. 5(a) which is an urban area surrounding the University of Pavia, Italy. It was recorded by the ROSIS-03 satellite sensor over an urban area surrounding the University of Pavia, Italy. It is of size 610 × 340 × 115 with a spatial resolution of 1.3 m/pixel and a spectral coverage ranging from

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TABLE I
CLASS LABELS OF PURDUE'S DATA

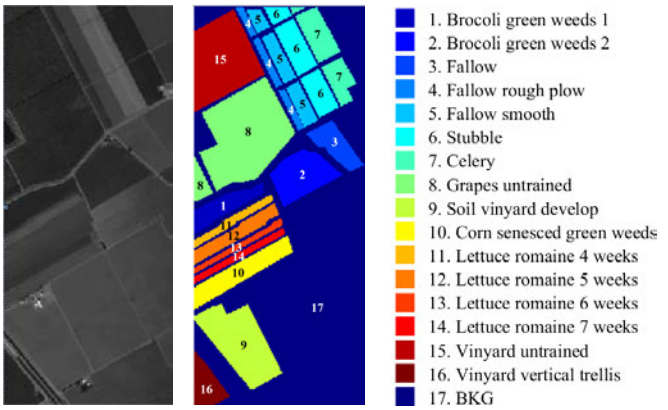
Class 1 (46)	Alfalfa	Class 7 (28)	Grass-pasture-mowed	Class 13 (205)	Wheat
Class 2 (1428)	Corn-notill	Class 8 (478)	Hay-windrowed	Class 14 (1265)	Woods
Class 3 (830)	Corn-mintill	Class 9 (20)	Oats	Class 15 (386)	Bldg-Grass-Trees-Drives
Class 4 (237)	Corn	Class 10 (972)	Soybean-notill	Class 16 (93)	Stone-Steel-Towers
Class 5 (483)	Grass-pasture	Class 11 (2455)	Soybean-mintill	Class 17 (10776)	BKG
Class 6 (730)	Grass-trees	Class 12 (593)	Soybean-clean		

TABLE II
CLASS LABELS OF CLASSES IN SALINAS

Class 1 (2009)	Brocoli green weeds 1	Class 10 (3278)	Corn senesced green weeds
Class 2 (3726)	Brocoli green weeds 2	Class 11 (1068)	Lettuce romaine 4 weeks
Class 3 (1976)	Fallow	Class 12 (1927)	Lettuce romaine 5 weeks
Class 4 (1394)	Fallow rough plow	Class 13 (916)	Lettuce romaine 6 weeks
Class 5 (2678)	Fallow smooth	Class 14 (1070)	Lettuce romaine 7 weeks
Class 6 (3959)	Stubble	Class 15 (7268)	Vinyard untrained
Class 7 (3579)	Celery	Class 16 (1807)	Vinyard vertical trellis
Class 8 (11271)	Grapes untrained	Class 17 (56975)	BKG
Class 9 (6203)	Soil vineyard develop		

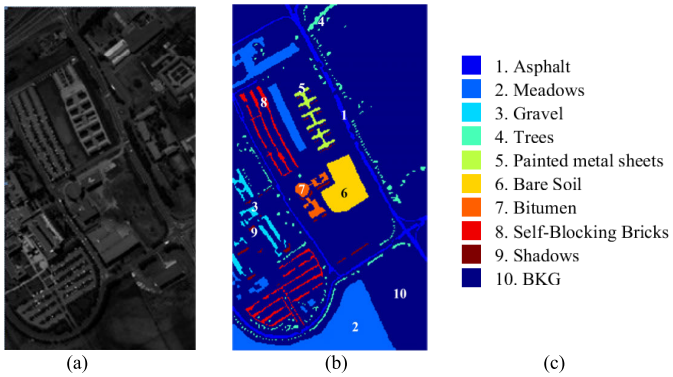
TABLE III
CLASS LABELS OF UNIVERSITY OF PAVIA

Class 1 (6631)	Asphalt	Class 5 (1345)	Painted metal sheets	Class 9 (947)	Shadows
Class 2 (18649)	Meadows	Class 6 (5029)	Bare Soil	Class 10 (164624)	BKG
Class 3 (2099)	Gravel	Class 7 (1330)	Bitumen		
Class 4 (3064)	Trees	Class 8 (3682)	Self-Blocking Bricks		



(a) (b) (c)

Fig. 4. Salinas and its GT with 16 classes. (a) Salinas scene. (b) Ground-truth image. (c) Classes by colors.



(a) (b) (c)

Fig. 5. University of Pavia and its GT with nine classes. (a) University of Pavia scene. (b) Ground-truth map. (c) Classes by colors.

0.43 to 0.86 μm with spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Fig. 5(b) provides its ground-truth map of nine classes along with color class labels in Fig. 5(c). Table III also tabulates the number of data samples in parentheses collected for each class. Like Salinas scene, this scene also has very large classes with only one small class with less than 1000 data samples, and class 9 with 947 samples. However, this scene has more complicated BKG than the other two studied scenes as already shown in 8 and 9 where the precision rate (PR) of this scene was much lower than the other two scenes. It should be noted that a total of 103 bands were used for experiments.

D. Houston Data

A fourth and final image scene is the Houston dataset, collected over the campus area of the University of Houston and its neighboring urban areas, as shown in Fig. 6(a) along with the GT of 15 classes in Fig. 6(b). The image was captured by a compact airborne spectrographic image (CASI) sensor in 2012 on June 23, 2012, between the times 17:37:10 to 17:39:50 UTC. The average height of the sensor above ground was 5500 ft. This dataset was made public in the 2013 IEEE GRSS Data Fusion Contest. It has dimensions of 349 $\times$ 1905 $\times$ 144, with 144 spectral bands ranging from 380 to 1050 nm and a spatial resolution of 2.5 m. Table IV provides a summary of the number of data samples, indicated in parentheses, collected for each class.

VII. CLASSIFICATION MEASURES FOR HSIC-B AND HSIC-NB

In order to measure the effectiveness of HSIC-NB, the commonly used OA and AA for HSIC-B are not

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TABLE IV
CLASS LABELS OF HOUSTON'S DATA

Class 1 (1251)	Healthy grass	Class 7 (1268)	Residential	Class 13 (469)	Parking lot 2
Class 2 (1254)	Stressed grass	Class 8 (1244)	Commercial	Class 14 (428)	Tennis court
Class 3 (697)	Synthetic grass	Class 9 (1252)	Road	Class 15 (660)	Running track
Class 4 (1244)	Tress	Class 10 (1227)	Highway	Class 16 (649816)	BKG
Class 5 (1242)	Soil	Class 11 (1235)	Railway		
Class 6 (325)	Water	Class 12 (1233)	Parking lot 1		

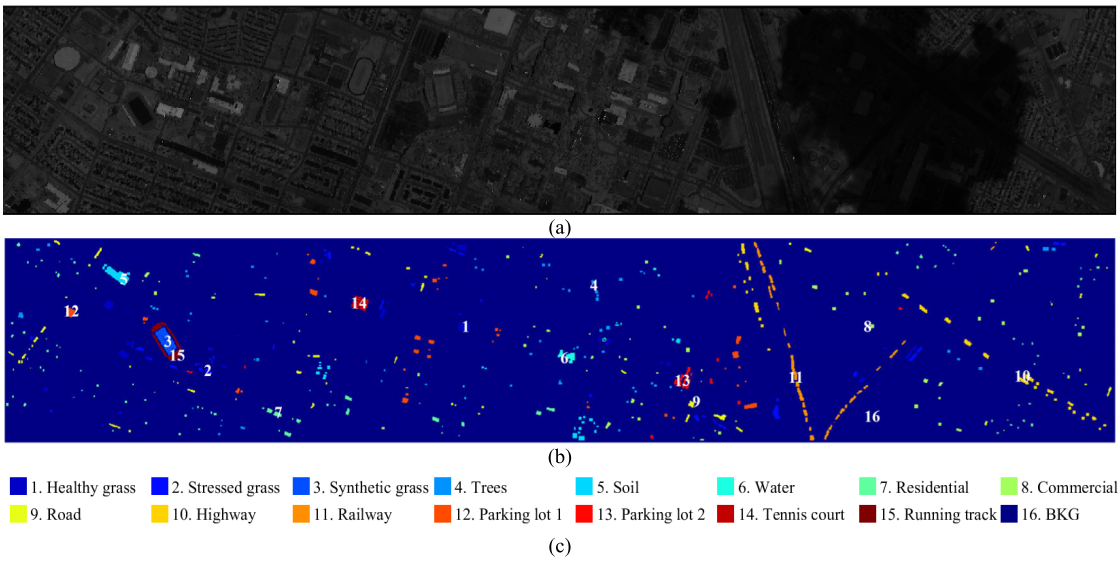


Fig. 6. Houston data and its GT with 15 classes, (a) Houston urban data; (b) ground truth; (c) 15 class labels.

applicable. From a statistical signal processing viewpoint [26, Section V], classification can be performed by two approaches, a priori classification with BKG removed by GT and a posteriori classification with BKG included.

A. A Priori Classification Measures for HSIC-NB

Assume that M is the number of classes, $\{C_j\}_{j=1}^{M}$, n_m is the number of data samples in C_m, and N is the total number of data samples, $N = \sum_{m=1}^{M} n_m$.

In the traditional HSIC-NB, the popular performance measurements are class accuracy (A) and OA, also known as OA is generally known as producer's accuracy given by

$$P_A(C_m) = \text{accuracy of the } m\text{th class} = \frac{n_{mm}}{\sum_{j=1}^{M} n_{jm}} = \frac{n_{mm}}{n_m} \tag{37}$$

and

$$P_{\text{OA}} = \frac{1}{N}\sum_{m=1}^{M} n_{mm} = \sum_{m=1}^{M} \frac{n_m}{N} P_A(C_m) \tag{38}$$

where (n_m/N) is the sample ratio of the mth class C_m. If we assume the probability of the mth class C_m to be equally likely by letting $(n_m/N) = (1/M)$, then (38) becomes AA, P_{AA}

$$P_{\text{AA}} = \frac{1}{M}\sum_{m=1}^{M} \frac{n_{mm}}{n_m} = \frac{1}{M}\sum_{m=1}^{M} P_A(C_m). \tag{39}$$

As noted in the introduction, the key difference between P_{OA} and P_{AA} lies in the fact that P_{OA} weights the accuracy of each class according to its sample size, while P_{AA} treats each class equally important. As a result, P_{OA} is generally dominated by the accuracies of large classes at the expense of the accuracies of small classes, while P_{AA} is not. Consequently, P_{AA} is more appropriate to deal with the imbalanced class issue.

B. A Posteriori Classification Measures for HSIC-B

The reason that P_{OA} in (38) and P_{AA} in (39) are referred to as a priori measures is because they assume that the complete GT is provided a priori so that the classification is performed with known BKG removed by GT. This common practice has been widely used in HSIC-NB in the literature. However, in many real scenarios, the complete GT may not be available and evaluating classification performance without BKG is not realistic. In this case, we need to factor BKG in classification evaluation. To cope with the issue, instead of a priori classification which calculates P_{OA} and P_{AA} for $\{x_n\}_{n=1}^{N}$, a posteriori classification measure is performed on the classified data samples, denoted by $\{\hat{x}_n\}_{n=1}^{N}$ according to classification maps, $\{\hat{C}_j\}_{j=1}^{M}$ produced by a classifier δ, in which case BKG is included for classification. In other words, a posteriori classification is carried out by designing an a posteriori classifier $\hat{\delta}$ that assigns a classified data sample $\hat{x}$ in $\hat{C}_m$ to a class C_m to which it is supposed to belong according to GT. In this case, we can further define a posteriori classification measures, $P_{\text{PR}}(\hat{C}_j)$ and PR, P_{PR}, corresponding to a priori classification measures, P_A in (37) and P_{OA} in (38)

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as follows:

$$P_{\mathrm{PR}}(\hat{C}_j) = \sum_{\mathbf{x}\in C_j, \hat{\mathbf{x}}\in \hat{C}_j} p\left(\mathbf{x}|\hat{\mathbf{x}}\right) = \frac{\hat{n}_{jj}}{\sum_{i=1}^{M} \hat{n}_{ji}} \tag{40}$$

where $\hat{n}_{ji}$ is the number of data samples classified in the jth class, C_j, which are supposed to be in the ith class, C_i and

$$P_{\mathrm{PR}} = \sum_{i=1}^{M} p(\hat{C}_i) P_{\mathrm{PR}}(\hat{C}_i) \tag{41}$$

where $p(\hat{C}_i) = (\hat{n}_i/N)$ is a posteriori probability of $\hat{C}_i$.

So, by virtue of (40) and (41), we can define the average PR (APR), P_{APR}, in correspondence to P_{AA} in (35) as

$$P_{\mathrm{APR}} = \frac{1}{M} \sum_{m=1}^{M} P_{\mathrm{PR}}(\hat{C}_m). \tag{42}$$

Since n_{mm} is identical to $\hat{n}_{mm}$, $\hat{N} = \sum_{i=1}^{M} \hat{n}_i$ is equal to $N = \sum_{i=1}^{M} n_i$. As a result, an overall PR (OPR), P_{OPR} in correspondence to P_{OA} in (38) can be further defined as

$$P_{\mathrm{OPR}} = \sum_{i=1}^{M} \left(\frac{\hat{n}_i}{\hat{N}}\right) P_{\mathrm{PR}}(\hat{C}_i). \tag{43}$$

Interestingly, if we remove BKG from classification, then

$$P_{\mathrm{OPR}} = \sum_{i=1}^{M} \left(\frac{\hat{n}_i}{\hat{N}}\right) P_{\mathrm{PR}}(\hat{C}_i) = \sum_{i=1}^{M} \left(\frac{n_i}{N}\right) P_A(C_i) = P_{\mathrm{OA}} \tag{44}$$

which implies that P_{OPR} is essentially the same P_{OA}.

It should be noted that PR is also known as user's accuracy to address BKG issue and is the one that can include all data samples for evaluation. For those who are interested in how to calculate OA, AA, and PR we refer details to the illustrative examples presented in [26, Section VI].

VIII. Experimental Results and Discussions

In order to evaluate the performance of HOCD, several classifiers were conducted for comparison in this section.

A. Compared HSIC Methods

1) OCC-Based Methods: The following four widely used OCC approaches were performed for comparison.

1) SVDD developed in [55] aims to find the minimum hypersphere enclosing the majority of the training samples.
2) KSR proposed in [57] maps the training samples into a high-dimensional feature space and builds a learning dictionary of each class for the sparse representation model, then optimizes the model to produce a reconstruction residual. The OCC results of each class are produced by binarizing the reconstruction residual maps.
3) FSSR [21] combines a band selection approach with an OCC classifier to improve OCC performance. FSSR first evaluates the reconstruction errors in the weight matrix obtained through sparse representation-based sample reconstruction. The spectral bands with lower reconstruction errors will be selected to perform OCC. To emphasize its use of OCC, FSSR to OCC-FSSR.
4) FSFR [21] evaluates the correlation among all bands using sparse representation. A band subset with low correlation with other bands is selected to implement OCC. It is also referred to as OCC-FSFR.

2) OCD-IRTS-Based Methods: In addition to the above four OCC approaches, IRTS-CEM [28], IRTS-KCEM [19], and IRTS-KTCIMF were also included for comparison since these methods were developed for HSIC-B. Nonetheless, it is worth noting that the original versions of IRTS-CEM and IRTS-KCEM were not directly applicable. First, the original versions of IRTS-CEM/KCEM performed M-class classification simultaneously, which is different from HOCD that conducts classification M times based on a hierarchical classification tree. Second, the final binarization maps $\{\mathrm{BMap}_m^{(k)}\}_{m=1}^{M}$ of IRTS-CEM/KCEM are produced by maximum a posteriori (MAP), which contradicts the HOCD method that uses the Otsu method. So, to make a fair comparison between IRTS-CEM/KCEM and HOCD-IRTS-KCEM/KTCIMF, we modified the original structure of IRTS-CEM/KCEM slightly. Instead of performing M-class classification simultaneously, IRTS-CEM/KCEM was performed class-by-class in a manner very similar to OCC, referred to as OCD-IRTS-CEM/KCEM. Then, the binarization map of each class was produced by Otsu's method instead of MAP used in IRTS-CEM/KCEM. In this way, OCD-IRTS-KCEM becomes a simplified version of the proposed HOCD-IRTS-KCEM method in the sense that OCD-IRTS-KCEM also performs OCD but without using CCP to prioritize each class and using $\mathbf{R}_{m-1}/P_{\mathbf{U}_{m-1}}^{\perp}$ to remove unwanted information. Therefore, three OCD-IRTS methods are used as baseline classifiers to evaluate the effectiveness of the proposed HOCD-KCEM/KTCIMF model.

3) IRTS-Based HSIC-B-Based Methods: In addition, two recent HSIC-B-based methods are also included for further comparison.

1) IRTS-3DCNN [20] which developed a new framework that can implement any arbitrary CNN as an initial classifier and then improve its performance through an iterative process.
2) IGLPN [21] which uses a Gaussian pyramid to implement the convolutional and pooling sublayers of a CNN and a Laplacian pyramid to capture additional multiscale features, while improving computational efficiency compared to the traditional CNN.

It is worth noting that in [8] and [9], both IRTS-3DCNN and IGPLN have shown to perform significantly better than most recent state-of-the-art HSIC-NB methods for both HSIC-NB and HSIC-B. So, in this article, only IRTS-3DCNN and IGPLN were implemented for benchmark comparison.

B. Parameter Setting

One key parameter used in HOCD is to utilize SID in (27) to measure similarity between the current class mean and means of previously classified classes. The tolerance value of τ used for SID in (27) for four datasets is listed in Table V. In addition, the parameter settings of IRTS-3DCNN and IGLPN used in [8] and [9] are also included in Table V.

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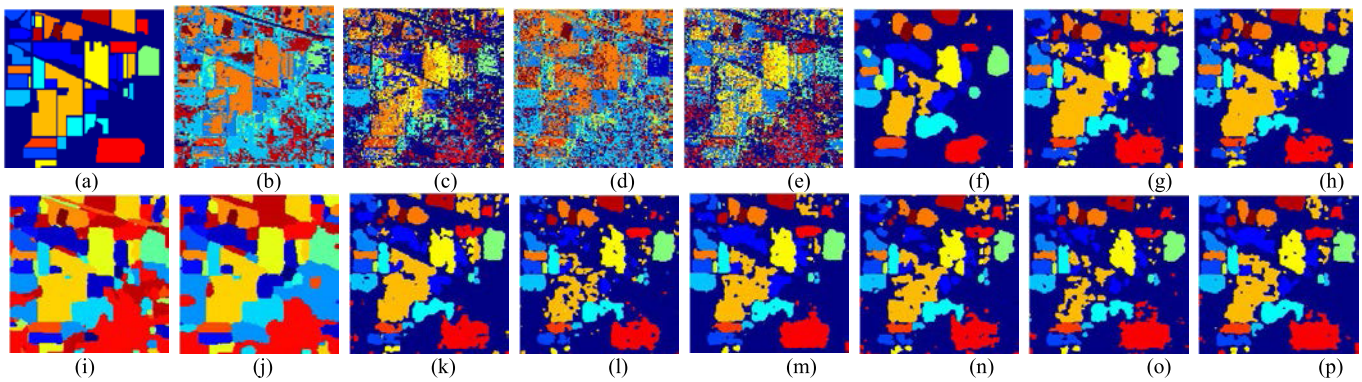


Fig. 7. Classification maps of Purdue data using 5% of training samples. (a) GT. (b) OCC-SVDD. (c) OCC-KSR. (d) OCC-FSSR. (e) OCC-FSFR. (f) OCD-IRTS-CEM. (g) OCD-IRTS-KCEM. (h) OCD-IRTS-KTCIMF. (i) IRTS-3DCNN. (j) IGLPN. (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM. (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM. (m) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM. (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF. (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF. (p) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KTCIMF.

TABLE V
PARAMETERS USED FOR PURDUE DATA, SALINAS, UNIVERSITY OF PAVIA, AND HOUSTON

HSI	τ of SID in (27)	(p,q) of IGLPN	nPC of IGLPN
Purdue	0.06	(3,2)	30
Salinas	0.8	(7,6)	15
U. of Pavia	10	(7,6)	15
Houston	0.34	(7,6)	15

One of the crucial issues in HSIC is how to select the training sample size for each of the classes to be classified. A commonly adopted approach to determine the size of training samples for each class is based on sample ratio. However, this selection often suffers from the so-called imbalanced class issue where a larger class has more training samples it needs, while a smaller class may have insufficient training samples.

To mitigate this issue, alternative methods have been proposed lately. One such method is proposed by Kang et al. [30] and Kang [31]. Another was developed by Zhong et al. [32], which utilizes a class feature-based approach to determine the training sample size for each class. Since Kang's method is available on website, this method was used in our experiments.

Table VI tabulates the training sample size of each class for the all four datasets used in our experiments where the data samples were randomly selected according to 1%, 5%, and 10% for Purdue and Houston; 0.5%, 1%, and 5% for Salinas and University of Pavia.

C. Purdue Indiana Indian Pines

Fig. 7 shows the classification maps with BKG produced by the 15 test methods with GT in Fig. 7(a). By visual inspection, it seems that the classification results clearly fall into three categories. The first category is made up of classification maps produced by OCC-based methods, (b) OCC-SVDD, (c) OCC-KSR, (d) OCC-FSSR, and (e) OCC-FSFR, which seemed to be worse ones because their classified data samples were distributed randomly like noise. A second category comprises classification maps produced by (i) IRTS-3DCNN and (j) IGLPN which look like segmentation results segmenting the entire image into 16 homogeneous regions. OCD-IRTS-based methods, a third category consists of classification maps produced by OCD-IRTS-based methods, (f) OCD-IRTS-CEM, (g) OCD-IRTS-KCEM, (h) OCD-IRTS-KTCIMF, and HOCD methods, (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM, (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM, (m) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM, (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF, (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF, and (p) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KTCIMF, which seemed among the best results since their classification results were more closer to the GT map in Fig. 7(b). Furthermore, if we compare the OCC-IRTS-CEM result in Fig. 7(f) to the classification results produced by CEM and TICMF using kernels in Fig. 7(g), (h), and (k)–(p), CEM had smooth class boundaries because CEM could not resolve nonlinear mixing on class boundaries but KCEM/KTCIMF could. As for classification maps produced by six HOCD-based methods, it is difficult to make a conclusion on which one is the best.

To quantitatively analyze the classification maps in Fig. 7, Table VII tabulates the results of P_{OA}, P_{AA}, $P_{\mathrm{OA}}(\mathrm{BKG})$, $P_{\mathrm{AA}}(\mathrm{BKG})$, $P_{\mathrm{OPR}}(\mathrm{BKG})$, and $P_{\mathrm{APR}}(\mathrm{BKG})$ produced by the 15 test methods using 5% training samples with best results are boldfaced where P_{OA} and P_{AA} represent accuracy excluding BKG, while $P_{\mathrm{OA}}(\mathrm{BKG})$ and $P_{\mathrm{AA}}(\mathrm{BKG})$ consider BKG as a separate class for classification. Specifically, P_{OPR} and P_{APR} are PR to account for BKG. Among these methods, $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM achieved the best precision in terms of $P_{\mathrm{OPR}}(\mathrm{BKG})$ and $P_{\mathrm{APR}}(\mathrm{BKG})$; $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM has the highest $P_{\mathrm{OA}}(\mathrm{BKG})$; IRTS-3DCNN achieved the best $P_{\mathrm{AA}}(\mathrm{BKG})$. However, IRTS-3DCNN's precision and $P_{\mathrm{OA}}(\mathrm{BKG})$ are very low because they classified every pixel. It is worth noting that $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM improved the result from OA/AA = 81.78/87.25%, OA(BKG)/AA(BKG) = 84.94/87.2% and OPR/APR = 84.79/81.22% to OA/AA = 82.05/88.92%, OA(BKG)/AA(BKG) = 85.88/88.96% and OPR/APR = 87.84/82.6% compared to OCC-IRTS-KCEM. $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-OCC-IRTS-KTCIMF also improved OA, AA, AA(BKG), and OPR compared to OCC-IRTS-KTCIMF.

There are several interesting observations and findings.

1) Interestingly, among the four OCC-based methods, the best one was OCC-SVDD which performed significantly better than the other three in terms of OA and AA.

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TABLE VI
TRAINING SAMPLE SIZES FOR PURDUE DATA, SALINAS, UNIVERSITY OF PAVIA, AND HOUSTON

Class	Purdue			Salinas			U. of Pavia			Houston		
	1%	5%	10%	0.5%	1%	5%	0.5%	1%	5%	1%	5%	10%
1	6	23	25	16	35	169	25	48	239	11	57	115
2	6	37	91	18	33	169	24	51	240	11	57	115
3	6	39	80	16	34	169	25	47	237	11	57	115
4	6	33	65	16	33	169	23	47	237	12	57	115
5	7	34	68	17	34	169	23	47	238	11	58	115
6	7	35	73	19	33	170	24	47	237	11	57	115
7	6	14	14	17	33	169	23	47	237	11	58	115
8	6	34	70	21	35	170	23	47	237	11	58	115
9	6	10	11	17	34	169	24	47	237	12	58	116
10	6	37	81	18	36	169				12	57	115
11	7	49	111	16	33	169				13	58	116
12	7	35	70	16	33	169				12	59	115
13	6	32	66	16	33	169				12	57	115
14	7	34	84	16	33	169				11	57	115
15	7	34	69	16	36	169				12	59	115
16	6	32	47	16	33	169						
Total	102	512	1025	271	541	2706	214	428	2139	173	864	1727

TABLE VII
P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{OPR}(BKG), AND P_{APR}(BKG) PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSSR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF USING 5% TRAINING SAMPLES FOR PURDUE DATA

	Method	P_{OA}/P_{AA} (%)	P_{OA}(BKG)/P_{AA}(BKG) (%)	P_{OPR}(BKG)/P_{APR}(BKG) (%)
OCC	OCC-SVDD	91.2/86.82	46.69/81.97	45.93/17.46
	OCC-KSR	64.54/56.46	62.48/56.7	55.08/40.94
	OCC-FSSR	71.31/74.95	40.49/71.2	37.29/17.42
	OCC-FSFR	62.28/58.13	52.93/57.3	42.99/24.61
OCD	OCD-IRTS-CEM	72.02/83.73	80.92/84.06	85.63/72.2
	OCD-IRTS-KCEM	81.78/87.25	84.94/87.29	84.79/81.22
	OCD-IRTS-KTCIMF	81.35/86.55	85.4/86.71	86.38/82.99
HSIC-NB	IRTS-3DCNN	98.56/99.26	48.04/**93.42**	48.04/48.82
	IGLPN	**98.9**/**98.95**	48.21/93.13	48.21/49.66
HOCD	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	81.86/87.73	84.99/87.74	86.14/81.21
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KCEM	65.28/81.72	78.93/82.31	**87.88**/**83.97**
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	82.05/88.92	**85.88**/88.96	87.84/82.6
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	77.34/84.92	82.8/85.1	85.78/81.9
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KTCIMF	69.47/83.47	80.21/83.88	87.02/83.59
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	81.84/88.89	85.23/88.87	86.93/82.85

However, OCC-SVDD did not do well on OA with BKG and AA with BKG. Nor did it on APR and OPR. Most interesting finding is that OCC-FSSR and OCC-FSFR did not do well regardless of whether or not BKG is included with OCC-FSFR being the worst one.

2) As for OCD-based methods, OCD-IRTS-CEM/KCEM/KTCIMF performed significantly better than the other four OCD-based methods.
3) As implemented for HSIC-NB, IRTS-3DCNN and IGPLN achieved the highest OA and AA values over 98.5% with BKG removed but very low OA and AA around 48% with BKG included. By contrast, ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-OCC-IRTS-KCEM achieved the highest OA and AA with BKG included at the expense of slightly low OA/AA without BKG. These results showed that no method can claim to do well both with/without BKG.
4) As expected, OCD/HOCD-based methods performed significantly better than OCC-based methods when BKG is included.

D. Salinas

Fig. 8 illustrates the classification maps with the BKG produced using 1% training samples by OCC-based methods, (b) OCC-SVDD, (c) OCC-KSR, (d) OCC-FSSR, (e) OCC-FSFR; OCD-IRTS-based methods, (f) OCD-IRTS-CEM, (g) OCD-IRTS-KCEM, (h) OCD-IRTS-KTCIMF; HSIC-NB-based methods, (i) IRTS-3DCNN, (j) IGLPN; HOCD-based methods, (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM, (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM, (m) ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM, (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF, (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF, and (p) ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF where the best results are boldfaced. It is very

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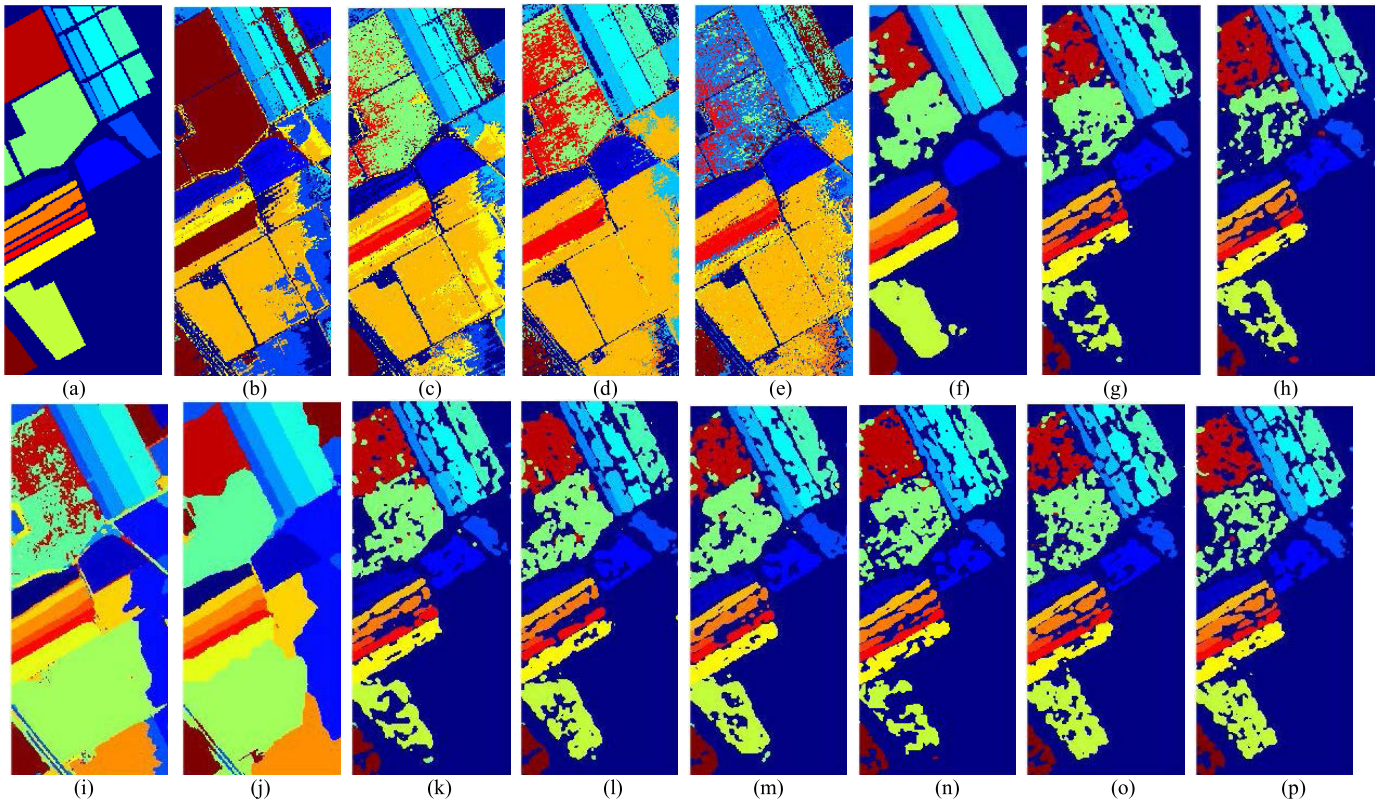


Fig. 8. Classification maps of Salinas data using 1% of training samples. (a) GT. (b) OCC-SVDD. (c) OCC-KSR. (d) OCC-FSSR. (e) OCC-FSFR. (f) OCD-IRTS-CEM. (g) OCD-IRTS-KCEM. (h) OCD-IRTS-KTCIMF. (i) IRTS-3DCNN. (j) IGLPN. (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM. (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM. (m) ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM. (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF. (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF. (p) ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF.

TABLE VIII

P_{OA}, P_{AA}, $P_{\mathrm{OA}}(\mathrm{BKG})$, $P_{\mathrm{AA}}(\mathrm{BKG})$, $P_{\mathrm{OPR}}(\mathrm{BKG})$, AND $P_{\mathrm{APR}}(\mathrm{BKG})$ PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSSR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF USING 1% TRAINING SAMPLES FOR SALINAS

	Method	$P_{\mathrm{OA}}/P_{\mathrm{AA}}$ (%)	$P_{\mathrm{OA}}(\mathrm{BKG})/P_{\mathrm{AA}}(\mathrm{BKG})$ (%)	$P_{\mathrm{OPR}}(\mathrm{BKG})/P_{\mathrm{APR}}(\mathrm{BKG})$ (%)
OCC	OCC-SVDD	92.52/92.45	55.82/88.25	51.08/38.88
	OCC-KSR	72.3/84.99	50.39/81.73	45.23/50.14
	OCC-FSSR	94.03/91.51	55.42/87.23	51.21/37.68
	OCC-FSFR	83.29/91.32	52.77/87.35	47.21/41.57
OCD	OCD-IRTS-CEM	86.55/91.05	**86.96**/90.83	85.07/79.06
	OCD-IRTS-KCEM	75.46/78.52	84.36/79.36	90.24/87.77
	OCD-IRTS-KTCIMF	74.56/78.48	83.86/79.31	90.08/87.66
HSIC	IRTS-3DCNN	94.38/97.36	45.98/91.64	45.98/52.97
	IGLPN	**99.32**/**99.52**	48.39/**93.66**	48.39/55.78
HOCD	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	78.16/79.38	85.3/80.13	89.78/86.95
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	79.08/78.94	85.71/79.71	89.44/86.18
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	79.54/79.79	85.71/80.48	88.88/85.69
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	74.04/76.48	84.09/77.49	**91.28**/**88.81**
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	74.83/77.14	84.35/78.09	91.13/88.62
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	75.18/78.24	84.42/79.12	90.85/88.48

clear that OCD- and HOCD-based methods produced the best results, while OCC produced the worst results. As for HSIC-NB methods, IGLPN seemed to perform better than IRTS-3DCNN.

Table VIII tabulates the results of P_A, P_{OA}, P_{AA}, $P_{\mathrm{OA}}(\mathrm{BKG})$, $P_{\mathrm{AA}}(\mathrm{BKG})$, $P_{\mathrm{OPR}}(\mathrm{BKG})$, and $P_{\mathrm{APR}}(\mathrm{BKG})$ produced by the 15 test methods using 1% training samples. P_{OA} and P_{AA} represent accuracy excluding BKG, while $P_{\mathrm{OA}}(\mathrm{BKG})$ and $P_{\mathrm{AA}}(\mathrm{BKG})$ consider BKG as a separate class for classification. P_{OPR} and P_{APR} are PR rates including BKG for classification. The best results are boldfaced.

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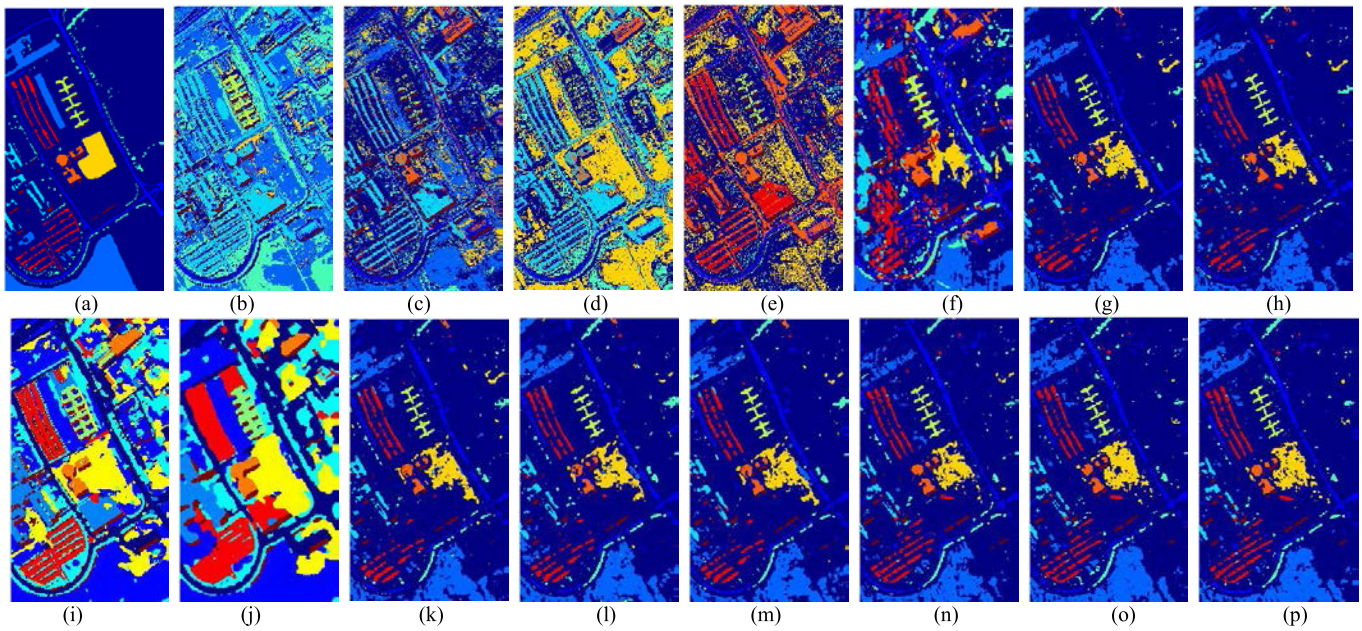


Fig. 9. Classification maps of University of Pavia using 1% of training samples. (a) GT. (b) OCC-SVDD. (c) OCC-KSR. (d) OCC-FSSR. (e) OCC-FSFR. (f) OCD-IRTS-CEM. (g) OCD-IRTS-KCEM. (h) OCD-IRTS-KTCIMF. (i) IRTS-3DCNN. (j) IGLPN. (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM. (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM. (m) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM. (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF. (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF. (p) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KTCIMF.

Among these methods, IGLPN achieved the best accuracy in terms of P_{OA}, $P_{\mathrm{AA},}$ and $P_{\mathrm{AA}}(\mathrm{BKG})$; OCC-IRTS-CEM achieved the best $P_{\mathrm{OA}}(\mathrm{BKG})$; HOCD-IRTS-KTCIMF($\mathbf{R}_m$) has the best precision in terms of $P_{\mathrm{OPR}}(\mathrm{BKG})$ and $P_{\mathrm{APR}}(\mathrm{BKG})$. It is worth noting that HOCD-IRTS-KTCIMF improved the PR rate compared to OCC-IRTS-KTCIMF, but it did not improve the accuracy as it did for Purdue data. In addition, HOCD-IRTS-KCEM only improved $P_{\mathrm{OA}}(\mathrm{BKG})$ and $P_{\mathrm{AA}}(\mathrm{BKG})$ compared to OCC-IRTS-KCEM. This is also different from the results of Purdue data.

E. University of Pavia

Fig. 9 shows the classification maps with the BKG produced using 1% training samples by OCC-based methods, (b) OCC-SVDD, (c) OCC-KSR, (d) OCC-FSSR, and (e) OCC-FSFR; OCD-IRTS-based methods, (f) OCD-IRTS-CEM, (g) OCD-IRTS-KCEM, and (h) OCD-IRTS-KTCIMF; HSIC-NB-based methods, (i) IRTS-3DCNN and (j) IGLPN; HOCD-based methods, (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM, (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM, (m) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM, (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF, (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF, and (p) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KTCIMF where HOCD-based methods produced nearly the same results by visual examination and seemed to be the best classifiers among all the test methods, whereas OCC-based methods yielded the worst results. Interestingly, the kernel versions of OCD-IRTS-based methods performed comparably to HOCD-based methods, while OCD-TGRS-CEM did not and encountered the issue with classes being heavily mixed. In analogy with the results of the Purdue and Salinas datasets, IRTS-3DCNN and IGPLN worked like segmentation techniques which segmented the entire image into nine class homogeneous regions.

To conduct a quantitative analysis, Table IX tabulates the results of P_A, P_{OA}, P_{AA}, $P_{\mathrm{OA}}(\mathrm{BKG})$, $P_{\mathrm{AA}}(\mathrm{BKG})$, P_{PR}, $P_{\mathrm{OPR}}(\mathrm{BKG})$, and $P_{\mathrm{APR}}(\mathrm{BKG})$ produced by the 15 test methods using 1% training samples. P_{OA} and P_{AA} represent accuracy excluding BKG, while $P_{\mathrm{OA}}(\mathrm{BKG})$ and $P_{\mathrm{AA}}(\mathrm{BKG})$ consider BKG as a separate class for classification. P_{OPR} and P_{APR} denote precision including BKG. The best results are boldfaced. Among these methods, $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KTCIMF has the best performance with the highest $P_{\mathrm{OA}}(\mathrm{BKG})$, $P_{\mathrm{OPR}}(\mathrm{BKG})$, and $P_{\mathrm{APR}}(\mathrm{BKG})$ as opposed to OCC-FSSR and OCC-FSFR which were the worst ones. Even though IRTS-3DCNN achieved the highest P_{AA} and $P_{\mathrm{AA}}(\mathrm{BKG})$, it has very low PR rates with only 20%. Although kernel versions of OCD-IRTS-based and HCOD-based methods performed relatively the same, HOCD-IRTS-KCEM did perform significantly better than OCD-IRTS-CEM in terms of PR rates at the expense of OA and AA.

F. Houston Data

Fig. 10 shows the classification maps with the BKG produced using 1% training samples by OCC-based methods, (b) OCC-SVDD, (c) OCC-KSR, (d) OCC-FSSR, and (e) OCC-FSFR; OCD-IRTS-based methods, (f) OCD-IRTS-CEM, (g) OCD-IRTS-KCEM, and (h) OCD-IRTS-KTCIMF; HSIC-NB-based methods, (i) IRTS-3DCNN and (j) IGLPN; and HOCD-based methods, (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM, (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM, (m) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KCEM, (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF, (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF, and (p) $(P^{\perp}_{\mathbf{U}_{m-1}}, \mathbf{R}_{m-1})$-HOCD-IRTS-KTCIMF. Once again, by visual inspection HOCD-based methods seemed to be the best among 15 test methods.

Since it is difficult to evaluate classification performance based on Fig. 10, Table X tabulates the results of P_A,

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TABLE IX

P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{OPR}(BKG), AND P_{APR}(BKG) PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSSR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF USING 1% TRAINING SAMPLES FOR UNIVERSITY OF PAVIA

	Method	P_{OA}/P_{AA} (%)	P_{OA}(BKG)/P_{AA}(BKG) (%)	P_{OPR}(BKG)/P_{APR}(BKG) (%)
OCC	OCC-SVDD	93.98/94.1	31.09/86.16	22.08/13.93
	OCC-KSR	72.97/76.14	61.18/74.34	30.58/25.11
	OCC-FSSR	58.21/66.92	42.79/64.11	18.11/38.27
	OCC-FSFR	56.51/70.7	55.85/69.2	23.24/32.07
OCD	OCD-IRTS-CEM	69.84/76.74	64.05/75.33	32.07/28.56
	OCD-IRTS-KCEM	61/66.53	84.57/68.95	62.73/62.12
	OCD-IRTS-KTCIMF	60.98/66.86	83.57/69.12	60.27/61.79
HSIC-NB	IRTS-3DCNN	98.23/**99.04**	20.26/**89.14**	20.26/22.23
	IGLPN	**98.92**/98.49	20.4/88.64	20.4/23.29
HOCD	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	60.88/66.8	84.64/69.2	63.08/61.9
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	64.7/67.87	85.05/70.12	63.21/62.61
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	60.94/67.6	84.27/69.87	61.94/61.66
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	60.46/68.12	85.04/70.44	64.62/64.05
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	64.2/69.63	84.89/71.7	63.03/63.35
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	63.89/69.11	**85.4**/71.3	**64.69**/**64.12**

TABLE X

P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{OPR}(BKG), AND P_{APR}(BKG) PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSSR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF USING 5% TRAINING SAMPLES FOR HOUSTON DATA

	Method	P_{OA}/P_{AA} (%)	P_{OA}(BKG)/P_{AA}(BKG) (%)	P_{OPR}(BKG)/P_{APR}(BKG) (%)
OCC	OCC-SVDD	96.55/96.6	4.92/90.72	2.57/0.54
	OCC-KSR	94.23/92.24	8.61/86.88	2.61/3.16
	OCC-FSSR	65.45/67.07	33.33/64.91	2.51/26.37
	OCC-FSFR	67.95/69.35	44.26/67.74	3.1/25.86
OCD	OCD-IRTS-CEM	69.51/73.02	76.3/73.23	7.28/25.28
	OCD-IRTS-KCEM	66.46/70.24	85.65/71.24	11.32/33.72
	OCD-IRTS-KTCIMF	58.87/63.95	88.54/65.53	12.76/32.73
HSIC-NB	IRTS-3DCNN	92.2/93.34	2.4/87.51	2.4/12.41
	IGLPN	**98.59**/**98.7**	2.56/**92.53**	2.56/9.73
HOCD	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	65.25/69.24	86.22/70.34	11.57/33.83
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	64.74/68.42	83.08/69.37	9.59/33.01
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	63.47/67.34	83.75/68.4	9.83/33.26
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	69.08/72.68	91.34/73.88	18.69/38.79
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	65.51/69.49	**91.75**/70.93	19.13/38.75
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	65.83/69.96	91.5/71.35	**19.2**/**39.07**

P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{PR}, P_{OPR}(BKG), and P_{APR}(BKG) produced by the 15 test methods using 1% training samples. P_{OA} and P_{AA} represent accuracy excluding BKG, while P_{OA}(BKG) and P_{AA}(BKG) consider BKG as a separate class for classification The best results are boldfaced. Due to very complicated BKG of this dataset, the APR and OPR rates were extremely low below, specifically, for OCC-based methods and IRTS-3DCNN and IGPLN with PR rates below 3% close to zero. Overall speaking, ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF yielded the best result to achieve the best P_{OPR}(BKG) and P_{APR}(BKG); $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF had the best P_{OA}(BKG); IGLPN had the best P_{OA}, P_{AA}, and P_{AA}(BKG). In addition, kernel versions of OCD-IRTS performed better than OCD-IRTS-CEM in terms of P_{OA}(BKG)/P_{AA}(BKG) but were worse than OCD-IRTS-CEM in terms of APR and OPR. Moreover, HOCD-IRTS-KTCIMF performed better than HOCD-IRTS-KCEM, particularly in improving both OA/AA and PR rates.

G. Comparative Analysis Using Various Training Sample Sizes

Since the size of training samples certainly affects classification performance, Tables XI–XIV summarize the results of P_A, P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{PR}, P_{OPR}(BKG), and P_{APR}(BKG) produced by the 15 test methods where P_{OA} and P_{AA} represent accuracy excluding BKG, while P_{OA}(BKG)

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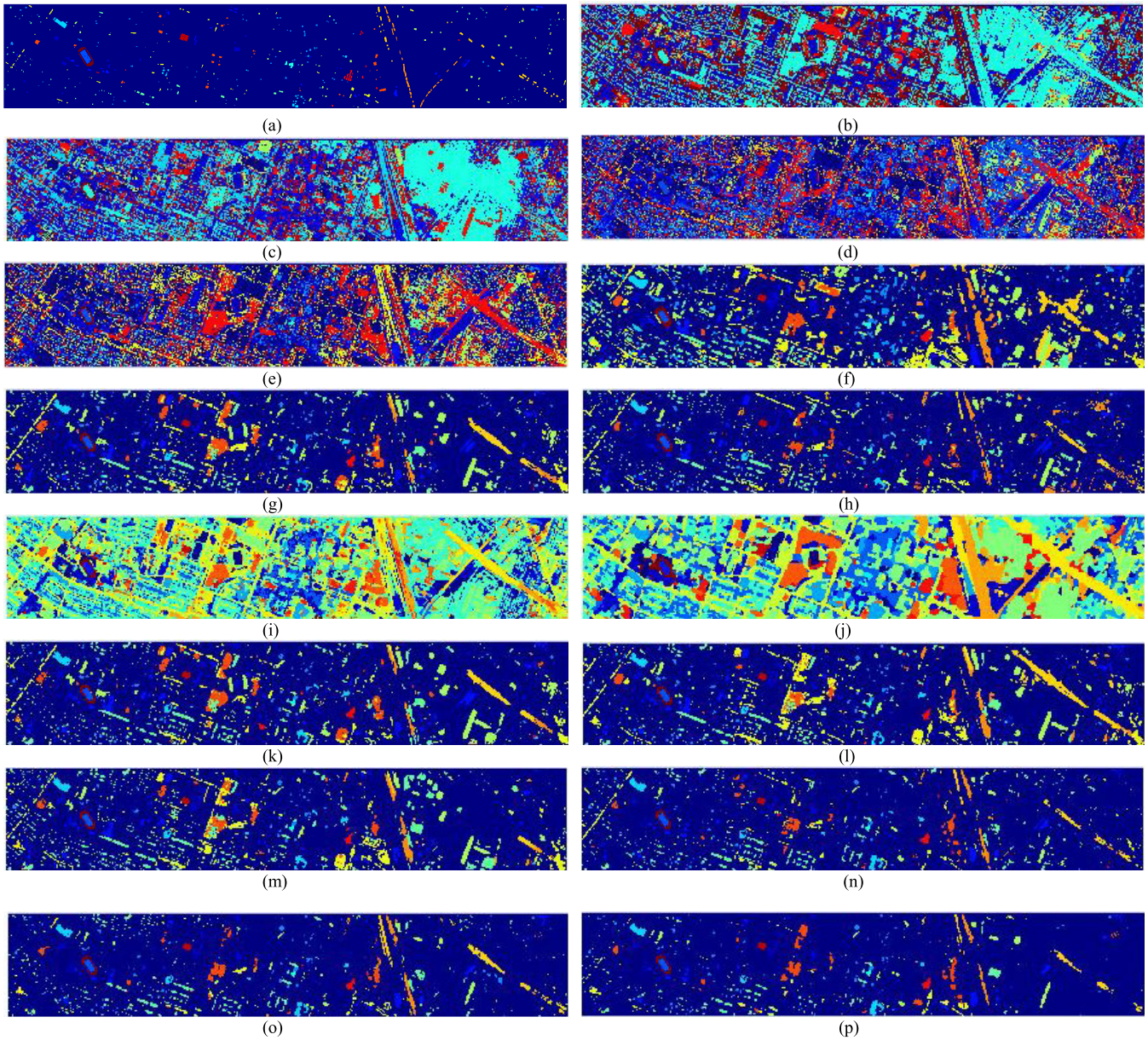


Fig. 10. Classification maps of Houston data using 5% of training samples. (a) GT. (b) OCC-SVDD. (c) OCC-KSR. (d) OCC-FSSR. (e) OCC-FSFR. (f) OCD-IRTS-CEM. (g) OCD-IRTS-KCEM. (h) OCD-IRTS-KTCIMF. (i) IRTS-3DCNN. (j) IGLPN. (k) $\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM. (l) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM. (m) ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM. (n) $\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF. (o) $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF. (p) ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF.

and P_{AA}(BKG) treat BKG as an additional and separate class for classification. Also, P_{OPR} and P_{APR} are PR rates by including BKG for classification. The best results are boldfaced. For the Purdue and Houston datasets, training sample sizes of 1%, 5%, and 10% were used and for Salinas and University of Pavia datasets, sample sizes of 0.5%, 1%, and 5% were used for study. For the Purdue data, OCD-based methods did not do well using 1% training samples. However, by using 5% and 10% training samples, OCD-based methods showed significant improvements. HOCD methods demonstrate better performance with larger training sample sizes. For Salinas, HOCD methods achieved better PR rates compared to OCD methods with greater improvements using higher training sample rates. Unlike the Purdue data, HOCD methods did not substantially improve classification accuracy for Salinas data. For the University of Pavia, comparing HOCD-IRTS-KCEM/KTCIMF methods to OCC-KCEM/KTCIMF, HOCD methods improved classification accuracy by using training samples larger than 1% and enhanced PR rates across various training sample sizes. For Houston, HOCD methods achieved the best P_{OPR} across different training sample rates. Although IGLPN achieved the best P_{AA}(BKG), it had very low PR rates. OCC-FSSR and OCC-FSSR also had high P_{APR}(BKG), but their classification accuracy and P_{OPR}(BKG) were very low. HOCD methods still performed better than these methods. These results indicate that HOCD methods generally improved performance, especially in terms of PR rates as the training sample size increases.

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TABLE XI

P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{OPR}(BKG), AND P_{APR}(BKG) PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSFR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF BY USING DIFFERENT TRAINING SAMPLES RATES FOR PURDUE DATA

Training rate %	Method	P_{OA}/P_{AA} (%)	$P_{OA}(BKG)/P_{AA}(BKG)$ (%)	$P_{OPR}(BKG)/P_{APR}(BKG)$ (%)
1%	OCC-SVDD	60.71/58.78	40.49/56.57	35.33/23.69
	OCC-KSR	43.3/39.58	41.39/39.58	30.72/17.18
	OCC-FSSR	40.65/41.19	33.95/40.39	25.32/39.78
	OCC-FSFR	13/18.43	45.09/21.8	23.28/49.51
	OCD-IRTS-CEM	49.47/65.99	62.55/66.52	**59.23**/67.8
	OCD-IRTS-KCEM	67.98/68.85	**68.48**/68.85	57.2/70.22
	OCD-IRTS-KTCIMF	66.23/67.01	68.36/67.2	57.48/70.95
	IRTS-3DCNN	87.21/94.06	42.51/88.53	42.51/45.05
	IGLPN	**90.17**/**95.56**	43.95/**89.94**	43.95/46.75
	$\mathbf{R}_{m\text{-}1}$-HOCD-IRTS-KCEM	70.47/71.92	67.75/71.52	56.8/68.87
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	56.82/67.67	49.78/66.22	39.98/64.83
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m\text{-}1}$)-HOCD-IRTS-KCEM	67.97/74.7	48.99/72.12	41.59/62.61
	$\mathbf{R}_{m\text{-}1}$-HOCD-IRTS-KTCIMF	62.32/64.31	65.02/64.5	54.35/**72.67**
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	54.07/64.61	56.35/64.26	45.67/68.7
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m\text{-}1}$)-HOCD-IRTS-KTCIMF	67.06/73	54.93/71.26	46.55/65.21
5%	OCC-SVDD	91.2/86.82	46.69/81.97	45.93/17.46
	OCC-KSR	64.54/56.46	62.48/56.7	55.08/40.94
	OCC-FSSR	71.31/74.95	40.49/71.2	37.29/17.42
	OCC-FSFR	62.28/58.13	52.93/57.3	42.99/24.61
	OCD-IRTS-CEM	72.02/83.73	80.92/84.06	85.63/72.2
	OCD-IRTS-KCEM	81.78/87.25	84.94/87.29	84.79/81.22
	OCD-IRTS-KTCIMF	81.35/86.55	85.4/86.71	86.38/82.99
	IRTS-3DCNN	98.56/**99.26**	48.04/**93.42**	48.04/48.82
	IGLPN	**98.9**/98.95	48.21/93.13	48.21/49.66
	$\mathbf{R}_{m\text{-}1}$-HOCD-IRTS-KCEM	81.86/87.73	84.99/87.74	86.14/81.21
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	65.28/81.72	78.93/82.31	**87.88**/**83.97**
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m\text{-}1}$)-HOCD-IRTS-KCEM	82.05/88.92	**85.88**/88.96	87.84/82.6
	$\mathbf{R}_{m\text{-}1}$-HOCD-IRTS-KTCIMF	77.34/84.92	82.8/85.1	85.78/81.9
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	69.47/83.47	80.21/83.88	87.02/83.59
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m\text{-}1}$)-HOCD-IRTS-KTCIMF	81.84/88.89	85.23/88.87	86.93/82.85
10%	OCC-SVDD	95.55/90.84	48/85.66	47.49/14.6
	OCC-KSR	70.43/59.41	61.86/59.07	54.79/45.7
	OCC-FSSR	89.74/86.7	48.14/82.11	46.13/16.5
	OCC-FSFR	70.38/69.03	45.8/66.28	40.31/25.96
	OCD-IRTS-CEM	79.68/88.12	83.65/88.08	85.26/70.92
	OCD-IRTS-KCEM	86.34/90.9	88.7/90.9	89.99/83.07
	OCD-IRTS-KTCIMF	86.33/91.08	88.63/91.07	89.81/83.06
	IRTS-3DCNN	98.89/**99.37**	48.21/**93.53**	48.21/50.29
	IGLPN	**99.4**/99.18	48.45/93.34	48.46/50.4
	$\mathbf{R}_{m\text{-}1}$-HOCD-IRTS-KCEM	87.15/91.61	88.66/91.52	89.2/82.31
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	85.81/91.24	88.76/91.26	**90.58**/**84.36**
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m\text{-}1}$)-HOCD-IRTS-KCEM	89.08/92.77	89.29/92.57	88.82/82.09
	$\mathbf{R}_{m\text{-}1}$-HOCD-IRTS-KTCIMF	87.37/91.22	88.92/91.17	89.58/82.77
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	88.22/92.33	89.57/92.24	90.15/83.98
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m\text{-}1}$)-HOCD-IRTS-KTCIMF	90.87/93.62	**90.06**/93.36	88.87/82.47

H. Discussions

As shown in the above experiments, a classification technique that performs very well on HSIC-NB is not necessary to work well on HSIC-B. To demonstrate this fact, four real popular hyperspectral images with various degrees of image complexity were selected for experiments. In addition, the 15 HSIC test methods to be compared were particularly selected to represent four categories.

1) *Category 1:* Four OCC-based methods—OCC-SVDD, OCC-KSR, OCC-FSSR, and OCC-FSFR.
2) *Category 2:* Three OCD-IRTS-based methods—OCD-IRTS-CEM, OCD-IRTS-KCEM, and OCD-IRTS-KTCIMF.

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TABLE XII

P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{OPR}(BKG), AND P_{APR}(BKG) PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSSR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF BY USING DIFFERENT TRAINING SAMPLE RATES FOR SALINAS

Training rate %	Method	P_{OA}/P_{AA} (%)	P_{OA}(BKG)/P_{AA}(BKG) (%)	P_{OPR}(BKG)/P_{APR}(BKG) (%)
0.5%	OCC-SVDD	85.7/85.58	56.51/82.24	51.04/45.36
	OCC-KSR	63.47/57	64.82/57.54	57.93/57.02
	OCC-FSSR	55.71/54.59	42.27/53.11	36.08/63.3
	OCC-FSFR	63.92/63.91	47.49/62.02	41.85/59.23
	OCD-IRTS-CEM	78.52/85.73	**84.29**/85.97	86.58/81.34
	OCD-IRTS-KCEM	70.43/72.31	82.21/73.55	89.58/86.93
	OCD-IRTS-KTCIMF	67.28/70.77	80.56/72.09	89.37/86.82
	IRTS-3DCNN	96.24/98.29	46.89/92.51	46.89/53.77
	IGLPN	**99.02**/**98.9**	48.24/**93.09**	48.24/55.8
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	68.96/70.96	81.59/72.29	89.65/87.39
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	68.09/70.59	80.38/71.85	87.67/85.3
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	69.41/72.25	80.82/73.39	87.12/84.91
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	66.89/68.87	80.7/70.34	**90.3**/**88.54**
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	65.27/68.76	79.91/70.24	88.86/87.13
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	65.02/66.68	79.87/68.29	89.24/87.56
1%	OCC-SVDD	92.52/92.45	55.82/88.25	51.08/38.88
	OCC-KSR	72.3/84.99	50.39/81.73	45.23/50.14
	OCC-FSSR	94.03/91.51	55.42/87.23	51.21/37.68
	OCC-FSFR	83.29/91.32	52.77/87.35	47.21/41.57
	OCD-IRTS-CEM	86.55/91.05	**86.96**/90.83	85.07/79.06
	OCD-IRTS-KCEM	75.46/78.52	84.36/79.36	90.24/87.77
	OCD-IRTS-KTCIMF	74.56/78.48	83.86/79.31	90.08/87.66
	IRTS-3DCNN	94.38/97.36	45.98/91.64	45.98/52.97
	IGLPN	**99.32**/**99.52**	48.39/**93.66**	48.39/55.78
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	78.16/79.38	85.3/80.13	89.78/86.95
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	79.08/78.94	85.71/79.71	89.44/86.18
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	79.54/79.79	85.71/80.48	88.88/85.69
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	74.04/76.48	84.09/77.49	**91.28**/**88.81**
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	74.83/77.14	84.35/78.09	91.13/88.62
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	75.18/78.24	84.42/79.12	90.85/88.48
5%	OCC-SVDD	98.33/98.37	51.59/93.01	49.79/27.99
	OCC-KSR	93.98/90.66	52.66/86.11	49.45/27.4
	OCC-FSSR	99.16/99.17	53.06/93.88	50.75/22.73
	OCC-FSFR	98.31/98.5	57.99/93.86	53.49/32.19
	OCD-IRTS-CEM	93.18/95.75	88.98/**95.12**	84.92/76.92
	OCD-IRTS-KCEM	92.05/94.67	92.86/94.61	93.19/90.41
	OCD-IRTS-KTCIMF	92.17/94.67	92.96/94.61	93.11/90.55
	IRTS-3DCNN	96.23/98.48	46.88/92.69	46.88/53.99
	IGLPN	**99.87**/**99.88**	48.65/94	48.65/56.76
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	92.46/94.85	93/94.77	93.07/90.29
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	92.33/94.17	92.83/94.12	92.79/89.49
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	93.2/94.54	**93.15**/94.46	92.65/89.13
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	91.5/94.3	92.94/94.3	**93.86**/**91.59**
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	90.76/93.79	92.52/93.81	93.68/91.48
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	89.52/92.99	91.94/93.06	93.63/91.48

3) *Category 3:* Two HSIC-NB methods—IRTS-3DCNN and IGPLN.
4) *Category 4:* Six HOCD-based methods—$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM/KTCIMF, $P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM/KTCIMF, and ($P^{\perp}_{\mathbf{U}_{m-1}}$, $\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM/KTCIMF.

In order to evaluate the performance of classification on an entire image with BKG included, the traditional P_{OA} and P_{AA} are not applicable. To address this issue, two types of measures are used for performance evaluation. One is to extend P_{OA} and P_{AA} to P_{OA}(BKG) and P_{AA}(BKG) which calculate P_{OA} and P_{AA} with BKG included as an additional separate class.

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TABLE XIII

P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{OPR}(BKG), AND P_{APR}(BKG) PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSSR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF BY USING DIFFERENT TRAINING SAMPLE RATES FOR UNIVERSITY OF PAVIA

Training rate %	Method	P_{OA}/P_{AA} (%)	P_{OA}(BKG)/P_{AA}(BKG) (%)	P_{OPR}(BKG)/P_{APR}(BKG) (%)
0.5%	OCC-SVDD	88.18/90.22	30.59/82.76	20.86/14.39
	OCC-KSR	79/71.6	37.3/67.08	21.35/24.17
	OCC-FSSR	75.56/75.8	43.08/71.68	22.21/26.68
	OCC-FSFR	65.87/63.4	55.54/62.34	25.79/29.58
	OCD-IRTS-CEM	60.33/70.78	60.51/69.76	27.66/27.39
	OCD-IRTS-KCEM	51.57/62.7	81.63/65.37	55.59/56.52
	OCD-IRTS-KTCIMF	53.8/62.6	81.83/65.25	55.94/56.78
	IRTS-3DCNN	97.2/**98.63**	20.05/**88.76**	20.05/20.99
	IGLPN	**98.25**/97.67	20.26/87.91	20.26/22.21
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	52.11/61.59	81.74/64.37	55.66/56.56
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	52.01/61.85	81.8/64.63	55.83/57.37
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	52.53/62.5	81.59/65.16	55.22/56.64
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	49.24/62.06	**82.87**/65.02	**59.89**/60.3
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	49.64/61.91	82.13/64.77	57.52/**60.37**
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{M-1}$)-HOCD-IRTS-KTCIMF	52.34/63.7	81.96/66.3	56.27/59.36
1%	OCC-SVDD	93.98/94.1	31.09/86.16	22.08/13.93
	OCC-KSR	72.97/76.14	61.18/74.34	30.58/25.11
	OCC-FSSR	58.21/66.92	42.79/64.11	18.11/38.27
	OCC-FSFR	56.51/70.7	55.85/69.2	23.24/32.07
	OCD-IRTS-CEM	69.84/76.74	64.05/75.33	32.07/28.56
	OCD-IRTS-KCEM	61/66.53	84.57/68.95	62.73/62.12
	OCD-IRTS-KTCIMF	60.98/66.86	83.57/69.12	60.27/61.79
	IRTS-3DCNN	98.23/**99.04**	20.26/**89.14**	20.26/22.23
	IGLPN	**98.92**/98.49	20.4/88.64	20.4/23.29
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	60.88/66.8	84.64/69.2	63.08/61.9
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	64.7/67.87	85.05/70.12	63.21/62.61
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	60.94/67.6	84.27/69.87	61.94/61.66
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	60.46/68.12	85.04/70.44	64.62/64.05
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	64.2/69.63	84.89/71.7	63.03/63.35
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	63.89/69.11	**85.4**/71.3	**64.69**/**64.12**
5%	OCC-SVDD	98.64/98.87	25.08/**89.58**	21.37/8.66
	OCC-KSR	90.53/85.61	45.9/80.48	26.1/19.09
	OCC-FSSR	98.01/97.62	31.4/89.27	22.83/15.89
	OCC-FSFR	96.21/96.02	37.42/88.64	24.22/17.45
	OCD-IRTS-CEM	78.37/83.17	62.29/80.67	32.29/26.81
	OCD-IRTS-KCEM	80.81/81.16	86.98/81.9	64.71/62.7
	OCD-IRTS-KTCIMF	80.35/80.41	86.89/81.23	64.52/62.92
	IRTS-3DCNN	98.69/**99.39**	20.35/89.45	20.36/22.52
	IGLPN	**99.32**/99.09	20.49/89.18	20.49/21.29
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	81.42/81.85	87.04/82.51	64.66/62.58
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KCEM	81.04/80.95	87.12/81.72	64.97/62.74
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	81.35/81.05	87.29/81.83	65.3/62.97
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	81.68/82.68	88.22/83.4	67.78/66.17
	$P^{\perp}_{\mathbf{U}_{m-1}}$-HOCD-IRTS-KTCIMF	81.44/82.81	**88.32**/83.54	**68.09**/**66.54**
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	81.3/82.96	88.14/83.65	67.63/66.21

Nonetheless, P_{OA}(BKG) and P_{AA}(BKG) are still a priori classification measures. Another is to calculate two PR rates, P_{APR} and P_{OPR} which are a posteriori measures based on classified data samples as opposed to the a priori measures P_{OA} and P_{AA} based on GT [26]. By virtue of P_{OA}(BKG), P_{AA}(BKG), P_{APR}, and P_{OPR} along with 15 test methods performed on four real datasets, the experimental results showed many interesting observation and findings summarized as follows.

1) OCC which was originally developed for OCC by treating a class of interest as a positive class and the remaining classes as a negative class did not perform well for HSIC-B. Among the four compared OCC-based methods,

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TABLE XIV

P_{OA}, P_{AA}, P_{OA}(BKG), P_{AA}(BKG), P_{OPR}(BKG), AND P_{APR}(BKG) PRODUCED BY OCC-SVDD, OCC-KSR, OCC-FSSR, OCC-FSFR, OCC-IRTS-CEM, OCC-IRTS-KCEM, OCC-IRTS-KTCIMF, IRTS-3DCNN, IGLPN, HOCD-IRTS-KCEM, AND HOCD-IRTS-KTCIMF BY USING DIFFERENT TRAINING SAMPLE RATES FOR HOUSTON

Training rate %	Method	P_{OA}/P_{AA} (%)	P_{OA}(BKG)/P_{AA}(BKG) (%)	P_{OPR}(BKG)/P_{APR}(BKG) (%)
1%	OCC-SVDD	82.04/81.99	9.95/77.37	2.32/5.18
	OCC-KSR	22.5/22.78	70.59/25.85	2.05/8.48
	OCC-FSSR	14.52/14.24	79.69/18.45	1.99/78.3
	OCC-FSFR	17.66/19.68	83.76/23.8	3.2/**81.03**
	OCD-IRTS-CEM	61.17/66.44	79.68/67.3	7.57/24.35
	OCD-IRTS-KCEM	54.72/59.01	87.24/60.83	11.11/32.47
	OCD-IRTS-KTCIMF	57.6/61.17	83.73/62.63	8.98/31.37
	IRTS-3DCNN	87.23/89.23	2.27/83.65	2.27/11.93
	IGLPN	**90.29**/**91.27**	2.35/**85.56**	2.35/9.26
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	54.78/58.82	87.04/60.64	10.81/33.01
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KCEM	57.25/60.7	82.95/62.13	8.58/32.01
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	57.07/60.62	83.21/62.07	8.65/31.27
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	56.97/61.78	91.17/63.68	**16.51**/36.54
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KTCIMF	54.26/59.77	**91.2**/61.8	16.24/36.99
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	56.86/61.63	90.87/63.51	16.08/36.54
5%	OCC-SVDD	96.55/96.6	4.92/90.72	2.57/0.54
	OCC-KSR	94.23/92.24	8.61/86.88	2.61/3.16
	OCC-FSSR	65.45/67.07	33.33/64.91	2.51/26.37
	OCC-FSFR	67.95/69.35	44.26/67.74	3.1/25.86
	OCC-IRTS-CEM	69.51/73.02	76.3/73.23	7.28/25.28
	OCC-IRTS-KCEM	66.46/70.24	85.65/71.24	11.32/33.72
	OCC-IRTS-KTCIMF	58.87/63.95	88.54/65.53	12.76/32.73
	IRTS-3DCNN	92.2/93.34	2.4/87.51	2.4/12.41
	IGLPN	**98.59**/**98.7**	2.56/**92.53**	2.56/9.73
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	65.25/69.24	86.22/70.34	11.57/33.83
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KCEM	64.74/68.42	83.08/69.37	9.59/33.01
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	63.47/67.34	83.75/68.4	9.83/33.26
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	69.08/72.68	91.34/73.88	18.69/38.79
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KTCIMF	65.51/69.49	**91.75**/70.93	19.13/38.75
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	65.83/69.96	91.5/71.35	**19.2**/**39.07**
10%	OCC-SVDD	98.34/98.41	5.03/92.42	2.61/0.43
	OCC-KSR	86.85/87.5	12.45/82.69	2.5/12.47
	OCC-FSSR	14.95/16.21	79.18/20.25	2.04/**89.34**
	OCC-FSFR	22.63/21.63	80.59/25.41	3.26/82.6
	OCD-IRTS-CEM	70.73/74.12	75.83/74.23	7.26/25.01
	OCD-IRTS-KCEM	70.95/73.78	84.53/74.48	11.11/32.51
	OCD-IRTS-KTCIMF	62.81/67.45	86.15/68.66	11.2/31.85
	IRTS-3DCNN	93.8/94.66	2.44/88.75	2.44/12.76
	IGLPN	**99.42**/**99.33**	2.58/**93.12**	2.58/8.89
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KCEM	69.92/72.94	84.21/73.67	10.79/32.23
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KCEM	68.97/72.1	83.35/72.83	10.14/32.53
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KCEM	68.1/71.44	83.8/72.24	10.39/32.81
	$\mathbf{R}_{m-1}$-HOCD-IRTS-KTCIMF	71.39/74.7	91.5/75.78	19.27/37.68
	$P^{\perp}_{\mathbf{U}_{m-1}}$ -HOCD-IRTS-KTCIMF	71.8/74.82	91.2/75.87	18.8/37.36
	($P^{\perp}_{\mathbf{U}_{m-1}}$,$\mathbf{R}_{m-1}$)-HOCD-IRTS-KTCIMF	69.99/73.29	**91.64**/74.47	**19.33**/37.65

OCC-SVDD was the best. The FSSR and FSFR shown to perform well in [21] turned out to perform worst on HSIC-B.

2) IRTS-3DCNN and IGPLN were shown to perform very well for HSIC-NB compared to most recent existing HSIC-NB-based methods including DL-based methods in [8] and [9]. However, it has also shown that HSIC-NB-based methods performed poorly for HSIC-B. This indicated that high classification accuracy for HSIC-NB is compromised by very low PR rates for HSIC-B. This is particularly true for DL-based methods which require estimate or reconstruct BKG via a training process

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when data BKG is very complicated such as Houston data [8], [9].

3) OCC is a priori classification as opposed to OCD which is a posteriori classification. As a result, OCC can do better than OCD in P_{OA} and P_{AA} compared to OCD which can do better than OCC in terms of PR.
4) Unlike OCC which still solves a binary classification problem, OCD is designed to solve a signal detection in noise problem for a classification problem where all classes other than the class of interest are considered as BKG or noise. With this interpretation, experiments demonstrated that OCD performed significantly better than OCC in terms of PR. This is also traded off for sacrificing high classification accuracy due to its conversion of a classification problem to a detection problem.
5) The classification results for HSIC-B produced by OCC-based methods look like detection maps with lots of noise scattering over the entire image, whereas the classification results for HSIC-B obtained by HSIC-NB-based methods look like segmentation maps which segment the entire image into a number of homogeneous regions. The classification results for HSIC-B derived by OCD/HOCD-based methods are indeed classification maps.
6) Since the classification in one layer produced by HOCD is seriously influenced by the classifications produced by previous layers, it is crucial to determine the orders of classes to be processed by HOCD. To resolve this issue, the hierarchical tree used by HOCD is specifically constructed to prioritize classes to be processed according to their priory scores calculated by CCP in (24).
7) The four datasets exhibit different degrees of image characteristics. PR provides a quantitative measure to evaluate the degree of complexity of BKG. By virtue of PR, Purdue and Salinas datasets have the same level of BKG complexity. This is followed by the University of Pavia which has next level of BKG complexity. The Houston data turns out to have the most complicated BKG.
8) According to the conducted experiments, there is no classification technique that can claim work for both HSIC-NB and HSIC-B. Apparently, if a classification technique performs well on HSIC-NB, it may not work well on HSIC-B, such as IRTS-3DCNN and IGPLN shown in [8] and [9]. In contrast, if a classification technique performs well on HSIC-B, it may not work well on HSIC-NB either. It seems that a technique cannot win both, taking good care of one issue at the expense of the other issue.
9) It should be noted that CEM produces a real-valued detection map. To perform detection, it requires a thresholding technique to obtain a binary detection map. Despite that many thresholding techniques have been proposed and suggested in the literature, Otsu's method remains one of the most widely used thresholding techniques and performs very effectively in various applications. As a matter of fact, Otsu's method used in HOCD has been shown to be satisfactory in HSIC [7], [8], [9], [27], [29], [45], hyperspectral target/anomaly detection [10], [11], [58], [59] as well as medical imaging detection [60]. So, its utilization in HOCD is verified and justified.
10) As for the computing efficiency of HOCD, IKCEM used in HOCD is a very efficient and effective detection technique. Its computational complexity is very simple. So, iterations do not really require much computing time. The computational cost is mainly resulting from its implementing kernels.
11) It should be noted that HOCD only needs tuning one key parameter, the threshold for SID in (27), which is used to remove low-similarity class information. Compared to many current DL methods which require tuning a large number of parameters, such as learning rates, regularization terms, and network architectures, HOCD remains relatively simple.
12) Finally, in order for readers to reproduce the results presented in this article, we make our codes available on website, https://umbc.atlassian.net/wiki/spaces/rssipl/pages/27885869/10.+Download.

IX. Discussions on HSIC-NB and HSIC-B

The experiments conducted in Sections VII-B to VII-E were designed to provide evidence that coping with HSIC-B is not trivial at all. As shown by experiments, a classification technique that performs very well on HSIC without BKG (HSIC-NB) is not necessary to work well on HSIC-B. This section discusses the differences between HSIC-NB and HSIC-B.

A. HSIC-NB

HSIC-NB is a generally training sampling-based classification method which makes use of training samples provided by GT to train classifiers or network models. This is particularly true for DL-based HSIC-NB methods which rely on training samples to fine tune a large number of parameters used to train their models. The reason that HSIC-NB methods can perform well is because GT is used to remove BKG samples from classification. As a result, BKG removal significantly reduced misclassification error caused by class boundary samples interacting BKG samples. Therefore, HSIC-NB methods can be considered as a priori classification according to [26]. However, when a complete GT is not available, BKG cannot be completely removed for classification, in which case HSIC-NB is no longer applicable and will break down. Most importantly, obtaining a creditable and reliable training sample set for BKG is challenging due to the fact that there are so many unknown and unlabeled data samples present in BKG that cannot be identified by visual inspection or prior knowledge. Under such circumstances, we must rely on HSIC-B.

B. HSIC-B

Unlike HSIC-NB which requires labeled samples to train classifiers or models, the proposed HOCD for HSIC-B is a detection technique which does not need training samples to train CEM used in HOCD. Instead, it only requires the knowledge of a desired target signature $\mathbf{d}$ for each class without knowing BKG. There are two ways to obtain $\mathbf{d}$. One

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is calculated by GT if it is available. The other is calculated by a selective set of training samples. As a consequence of using CEM, HOCD can perform well on images including BKG because CEM can classify a class of interest as a desired signature **d** for target detection while suppressing all data samples other than the class of interest as BKG samples.

In order to evaluate the performance of classification on an entire image with BKG, the traditional P_{OA} and P_{AA} used for HSIC-NB are not applicable and must be extended to P_{OA}(BKG) and P_{AA}(BKG) which calculate P_{OA} and P_{AA} with BKG included as an additional separate class. Another two PR rates, P_{APR} and P_{OPR}, which are a posteriori measures based on classified data samples proposed in [26] as opposed to P_{OA} and P_{AA} as the a priori measures based on GT can be also used for HSIC-B. So, in this case, HSIC-B can be considered as a posteriori classification [26].

It is worth noting that if HSIC-B is used as HSIC-NB for classification with BKG removed, it does not perform as well as HSIC-NB does shown by the experiments in Section VIII. This makes sense since HSIC-B does not take advantage of GT as HSIC-NB does to remove BKG from classification.

X. Conclusion

A major strength of HSI is to uncover many unknown substances that cannot be identified by prior knowledge or visual examination. Accordingly, how to cope with this issue is very challenging. Unfortunately, most current research in HSIC has been mainly focused on HSIC-B removed by GT. This is an unrealistic scenario because it requires a complete GT to remove BKG. Although many DL approaches have been proposed to estimate or reconstruct BKG for BKG removal, finding creditable training samples from unlabeled BKG still remains unsolved. This article attempts to address an important issue in HSIC which has not been investigated in the past but is very challenging in real practical applications, that is, how to classify an entire hyperspectral image including BKG. Specifically, it presents a signal detection approach with no need of BKG training samples to perform HSIC-B included. It is called HOCD, which implements OCD via a hierarchical tree, to address the BKG issue.

The novelties of this article can be summarized as follows.

1) The key novelty is to develop an OCD approach that converts an HSIC-B problem to a signal detection problem in BKG.
2) Instead of traditional classifiers used in HSIC, a subpixel target detection, IKCEM or IKTCIMF is used in OCD.
3) Because OCD is carried out by one class at a time, for OCD to perform multiclass classification, OCD is implemented in a hierarchical tree which detects one class at a time in one layer.
4) Since any class can be classified in any layer in HOCD, HOCD requires a ranking of all classes so that each class will be classified according to its significance. To meet this need, a new concept of CCP derived from CEM is specifically designed to account for class prioritization.

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